

Unblinding Request: 2021 Full Luminosity

Emrys Peets

Stanford University
SLAC National Accelerator Laboratory



Stanford
University



NATIONAL
ACCELERATOR
LABORATORY

Unblinding 2021 Checklist

- I. Finalize Event Selection
- II. Background Model Validation
 - A. Quality Fits/Signal Extraction
 - B. Background Profiling
 - C. Viable Search Window for 2021
- III. Signal Model Choices
 - A. Window Range, signal template
- IV. Discovery Framework
 - A. Asymptotic Likelihood
- V. No siggie? No Biggie! – upper limits
 - A. CLs to determine signal yield in individual datasets
 - B. CLs to determine shared coupling when combining datasets

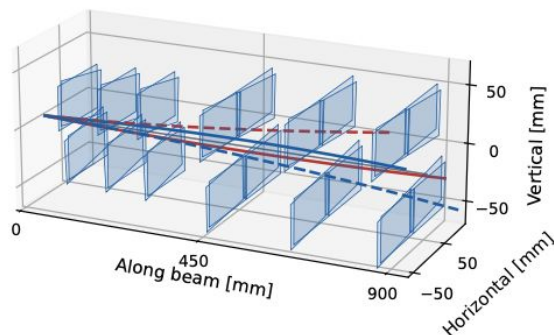
Results may be independent of coupling strength → We are looking at peaking structure over a QED background. This can be input to a variety of frameworks. Asymptotic probabilities and independent experiment hypotheses should be published and considered meaningful.

Event Selection (2021) (Pre-Selection+)

[put a table on event selection here and a few of the representative plots of kinematic distributions on this slide from v5.0.5 note - new is 8 hit electron and removing vertex χ^2 requirement]

Acceptances by Geometry

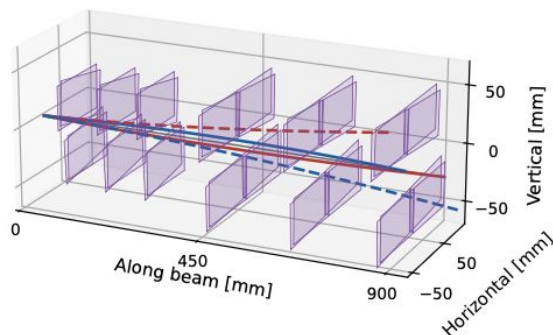
HPS 2015 | 1.056 GeV



Passing examples: 20 and 195 MeV

Nominal 15 mrad scale ≈ 15.84 MeV

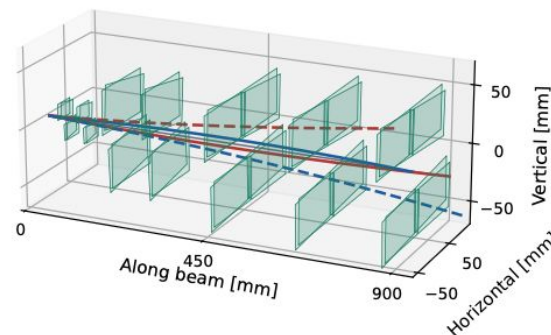
HPS 2016 | 2.300 GeV



Passing examples: 40 and 425 MeV

Nominal 15 mrad scale ≈ 34.50 MeV

HPS 2021 | 3.740 GeV



Passing examples: 60 and 715 MeV

Nominal 15 mrad scale ≈ 56.10 MeV

— Positron — Electron — Low-mass example - - - High-mass example

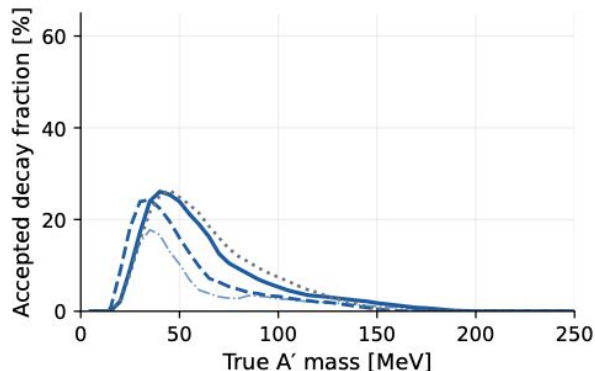
Note goes into acceptances more thoroughly, will start search in 2021 100% at 56 MeV

Acceptances by Campaign

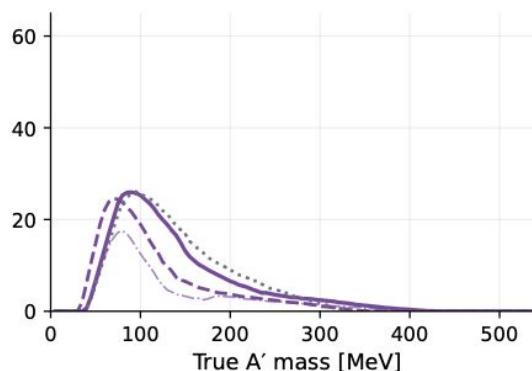
Run	$B_y(\mathbf{r}_{\text{map},0})$ [T]	Sampled masses [MeV]	Peak [MeV]	Peak fraction
2015	-0.2402	20–195	40	26.1%
2016	-0.5234	40–425	90	25.9%
2021	-0.8596	60–715	195	40.1%

— Field + hit cuts, $x = 1$ B = 0 + hit cuts, $x = 1$
--- Field + hit cuts, $x = 0.8$ -.-.- Field + all stations, $x = 1$

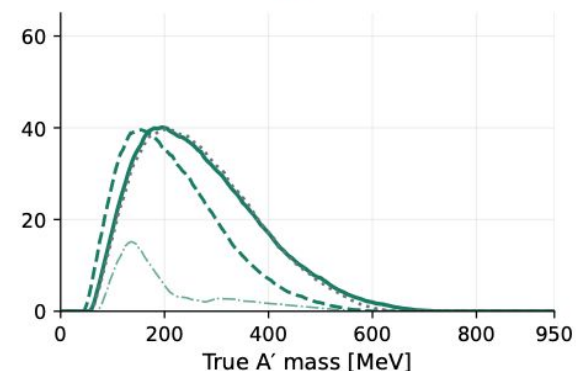
HPS 2015 | 1.056 GeV



HPS 2016 | 2.300 GeV

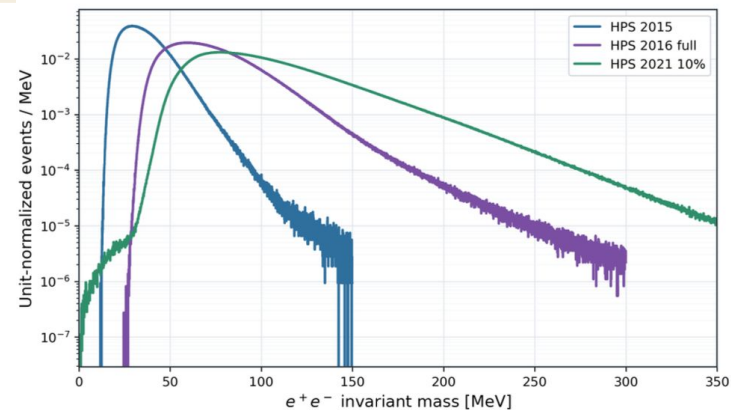


HPS 2021 | 3.740 GeV

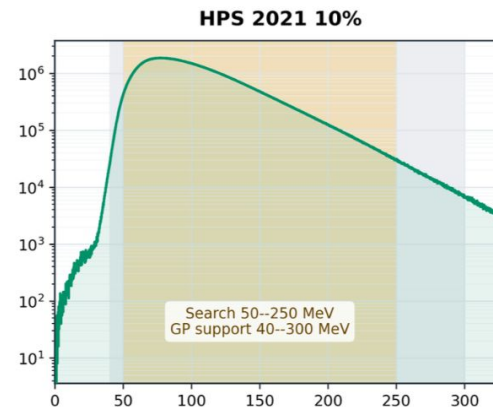
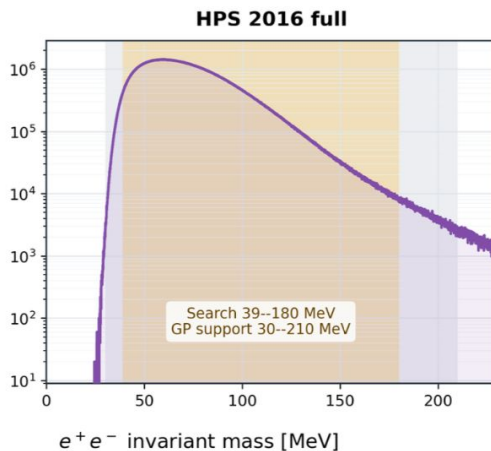
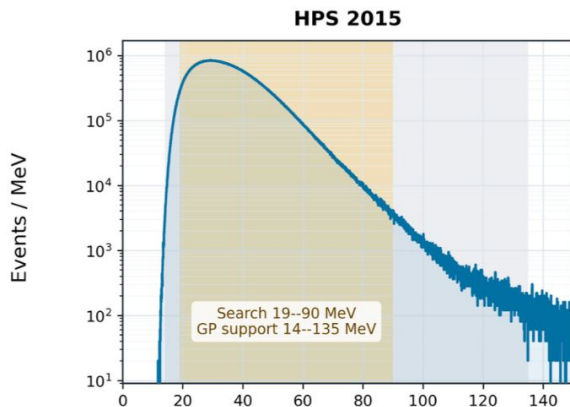


HPS datasets for this phase

Run period	Beam energy	Sample used here	Search interval	GP fit support
2015 engineering run	1.056 GeV	1170 nb ⁻¹	19 MeV to 90 MeV	14 MeV to 135 MeV
2016 engineering run	2.3 GeV	10 608 nb ⁻¹	39 MeV to 180 MeV	30 MeV to 210 MeV
2021 physics run	3.74 GeV	10% distributed development subset of a $\sim 160 \text{ pb}^{-1}$ parent run	50 MeV to 250 MeV	40 MeV to 300 MeV



Gray: GP fit support Gold: tested signal-search interval



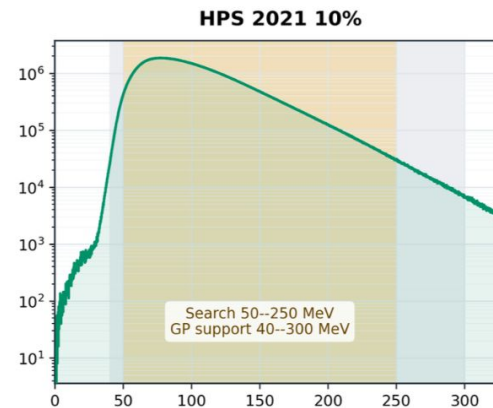
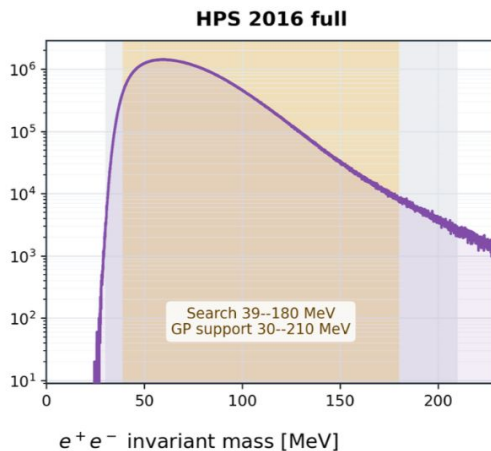
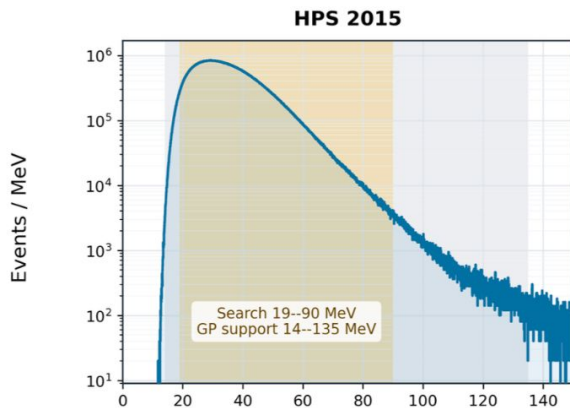
HPS datasets for this phase

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new fit support for 2021 $\rightarrow$ 36 MeV
2015 Search Tentatively Extended $\rightarrow$ 100 MeV

[make this slide and the previous slide have plots with the new fit support and 2015 search range]

Gray: GP fit support Gold: tested signal-search interval



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[i want this slide to explain/illustrate length scale in one figure, Constant in one figure, and alpha in one figure, can look through various other presentations to get this info or make new ones to easily explain to people]

$$k(x, x') = C \exp \left[-\frac{(x - x')^2}{2\ell^2} \right]$$

Background Profiling



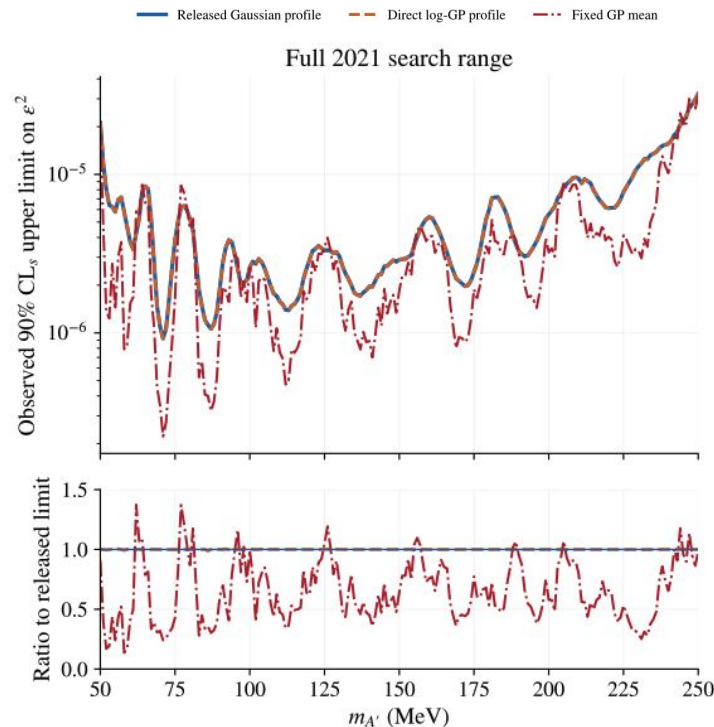
local Poisson mean

$$\lambda_d(A_d, \theta_d) = \underbrace{b_d}_{\text{projected bkg}} + \underbrace{L_d \theta_d}_{\text{profiled bkg}} + \underbrace{A_d w_d}_{\text{signal yield * template}}$$

profiled local likelihood

$$\mathcal{L}_d(A_d, \theta_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}(n_{d,i} | b_{d,i} + (L_d \theta_d)_i + A_d w_{d,i}(m)) \right] \exp\left(-\frac{1}{2} \theta_d^T \theta_d\right)$$

2021 10%: background profiling comparison



Background Profiling



local Poisson mean

$$\lambda_d(A_d, \theta_d) = \underbrace{b_d}_{\text{projected bkg}} + \underbrace{L_d \theta_d}_{\text{profiled bkg}} + \underbrace{A_d w_d}_{\text{signal yield} * \text{template}}$$

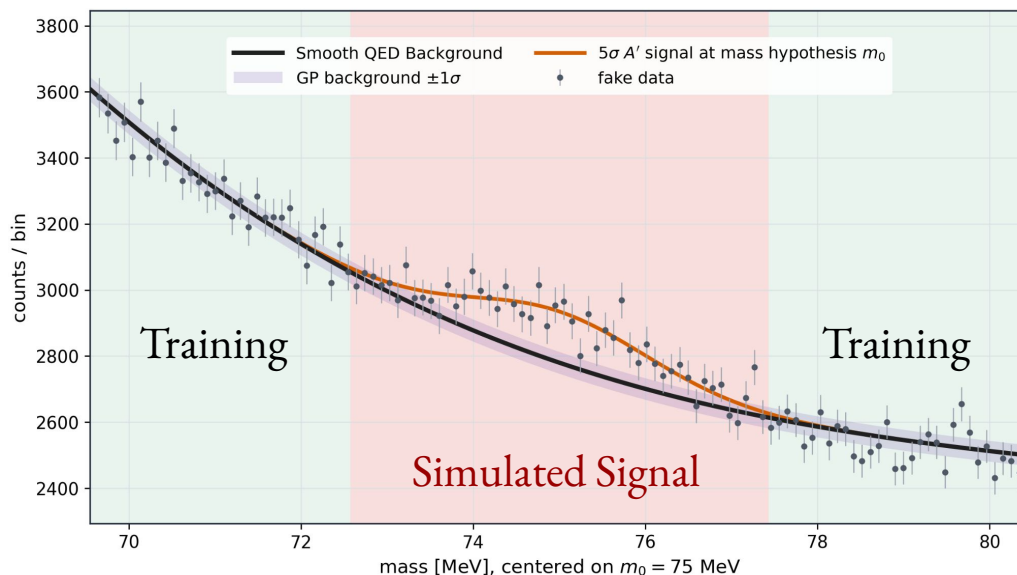
profiled local likelihood

$$\mathcal{L}_d(A_d, \theta_d) = \left[\prod_{i=1}^{N_d(m)} \text{Pois}(n_{d,i} | b_{d,i} + (L_d \theta_d)_i + A_d w_{d,i}(m)) \right] \exp\left(-\frac{1}{2} \theta_d^\top \theta_d\right)$$

[place good background profiled plot from a signal extraction plot from 2021 in this location]

Prompt A' Signal Model and Backgrounds

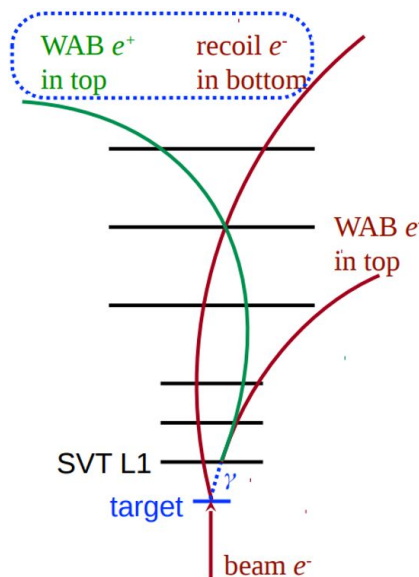
If A' exists within the acceptance of HPS, it will present itself as a **gaussian excess above background** in the IMD.



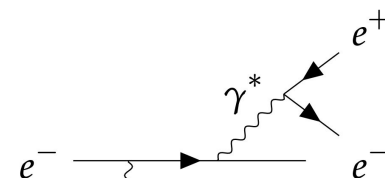
natural width of A' $\ll$ detector resolution
observed signal width = experimental mass resolution

Primary Backgrounds

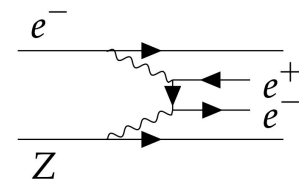
Wide Angle Brem



Radiative tridents



Bethe-Heitler tridents



Fitting the invariant mass distribution pt1

[show a few fits and also χ^2/dof in 2021 10% across invariant mass, imagine this information is stored, maybe KS χ^2 as well but make sure each of these pts are meaningful and aren't too wordy]

Fitting the invariant mass distribution pt2

[show a few fits and also χ^2/dof in 2021 10% across invariant mass, imagine this information is stored, maybe KS χ^2 as well but make sure each of these pts are meaningful and aren't too wordy]

2021 functional form **fShiftSigPowTail** (excluding 65 MeV region)

$$B_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\frac{m - m_{0,d}}{\theta_d} + c_{1,d} u_d + c_{2,d} u_d^2 \right]$$

$$S(m; t, w) = [1 + \exp(-(m - t)/w)]^{-1} \quad u_d = (m - m_{\text{mid},d})/\tilde{m}_{\text{scale},d}$$

2015/2016 functional form **fGenGammaThresh**

$$G_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\left(\frac{m - m_{0,d}}{\lambda_d} \right)^{p_d} \right]$$

2021 functional form **fShiftSigPowTail** (excluding 65 MeV region)

$$B_d(m) = N_d S(m; t_d, w_d) (m - m_{0,d})^{a_d} \exp \left[-\frac{m - m_{0,d}}{\theta_d} + c_{1,d} u_d + c_{2,d} u_d^2 \right]$$

$$S(m; t, w) = [1 + \exp(-(m - t)/w)]^{-1} \quad u_d = (m - m_{\text{mid},d})/\tilde{m}_{\text{scale},d}$$

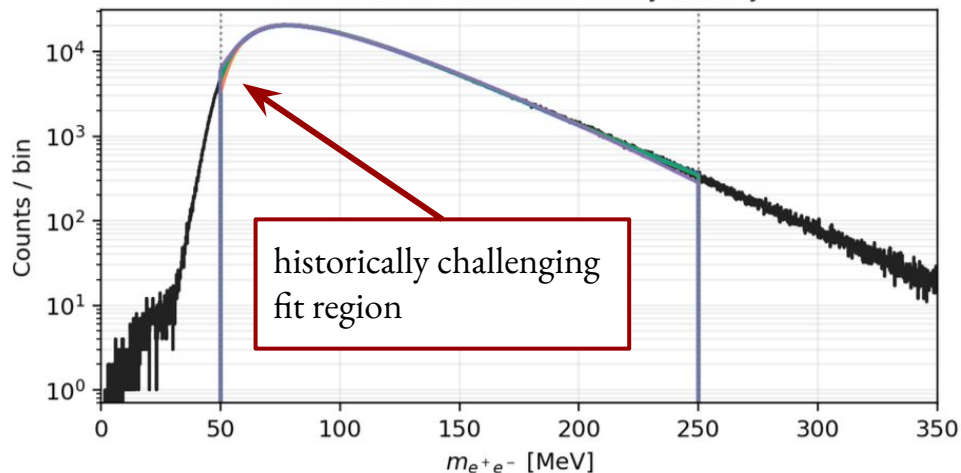
Four different statistical validation tests conducted

- I. 1% Fit * 10 (to double check 10% decisions)
- II. 1% Fit * 100 (to predict 100%)
- III. 10% Fit (to validate modeling on 10% data compare with scaled 1%)
- IV. 10% Fit * 10 (to predict 100% and compare with scaled 1%)

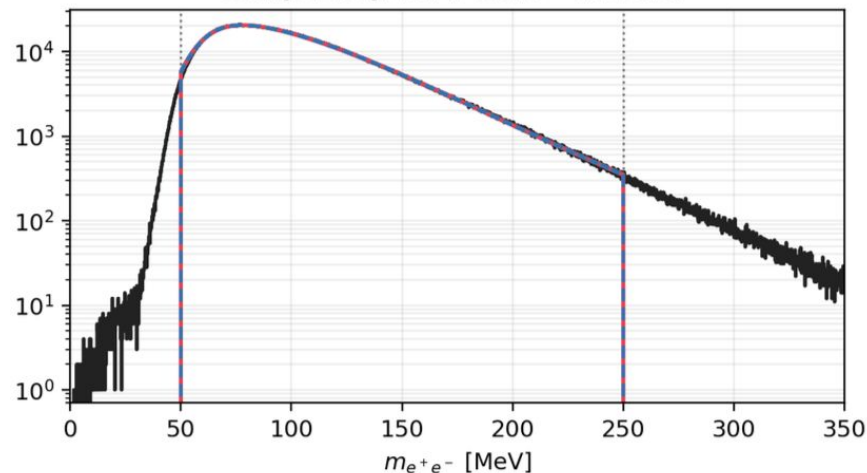
Example Fits to 1% Dataset

- Input histogram
- sigmoid*power*exp + raw expquad ($\chi^2/\text{ndof}=1.17$)
- thresholded gen-gamma ($\chi^2/\text{ndof}=1.07$)
- shifted gen-gamma ($\chi^2/\text{ndof}=1.19$)
- endpoint-aware sigmoid*power*exp ($\chi^2/\text{ndof}=3.75$)

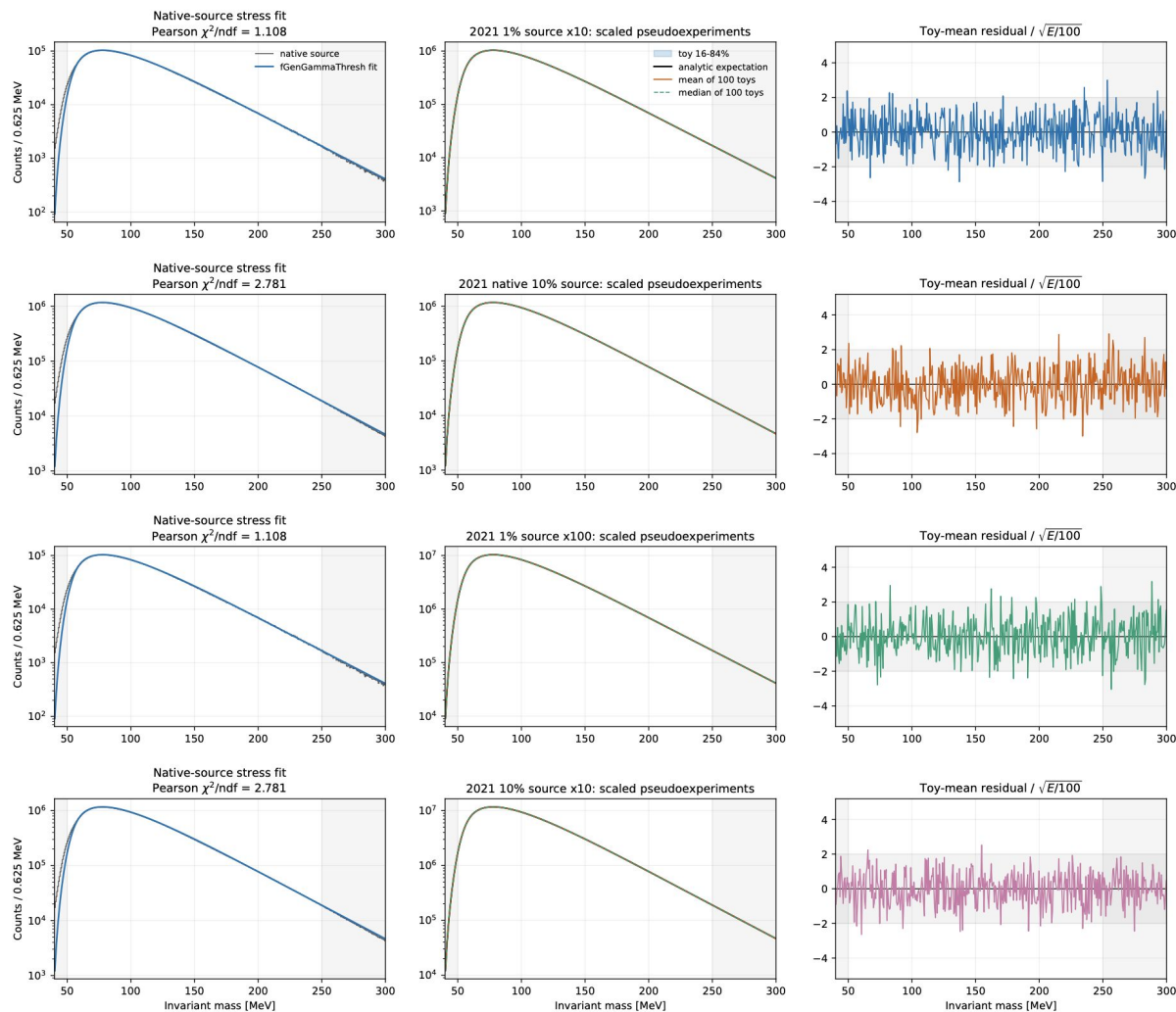
HPS 2021 1% candidate-family overlay



HPS 2021 1% / sigmoid*power*exp + raw expquad
Scan [50, 250] MeV / N events = 1.229×10^7



Pseudoexperiments



Validation Metrics

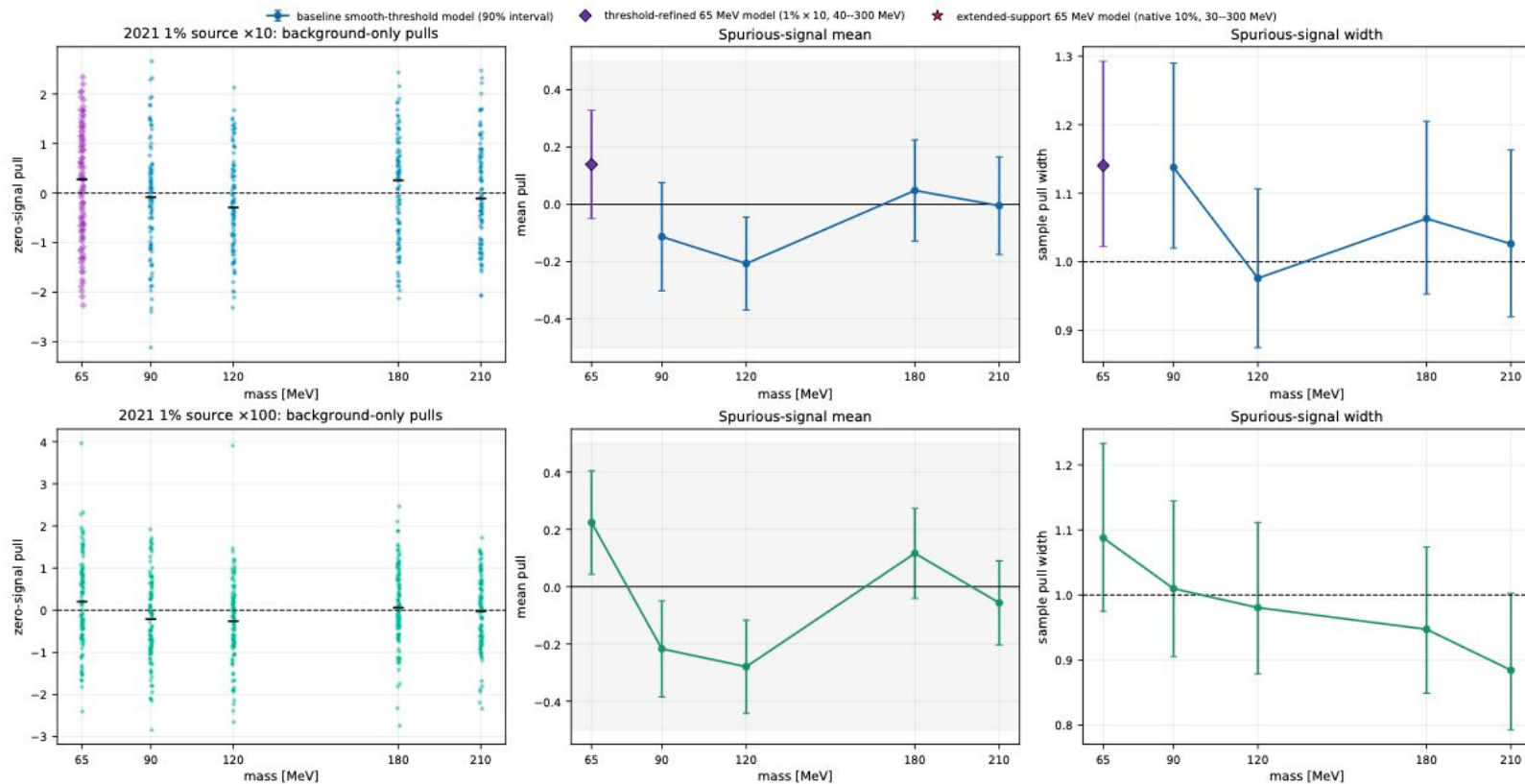
[explain briefly importance]

- Pull Mean [insert exact equation]
- Pull Width

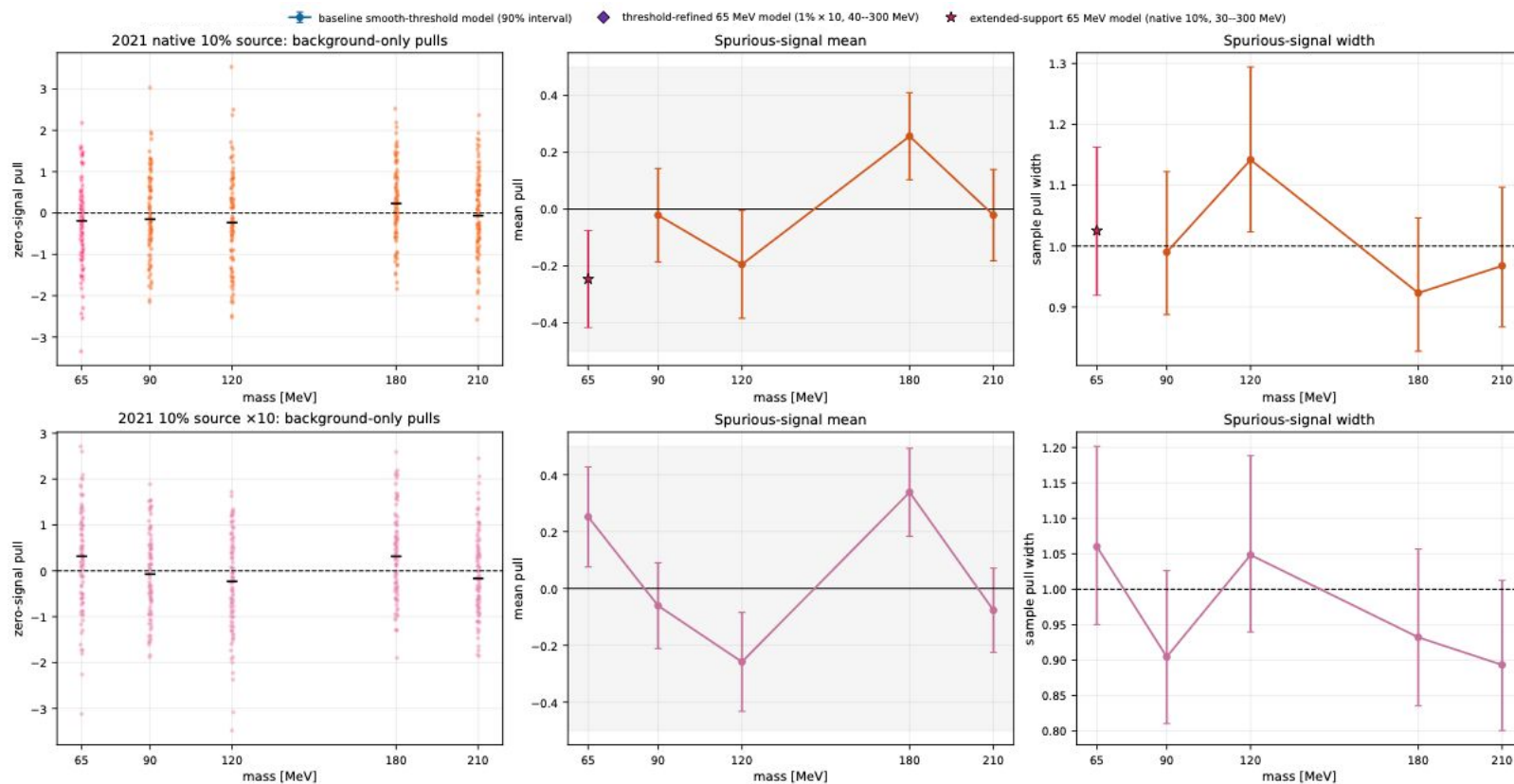
Studies Conducted

- I. Spurious Signal (0 injection)
- II. Injection Extraction (1,3,5 Sigma)

BKG Only Pseudoexperiments (1% Source)



BKG Only Pseudoexperiments (10% Source)



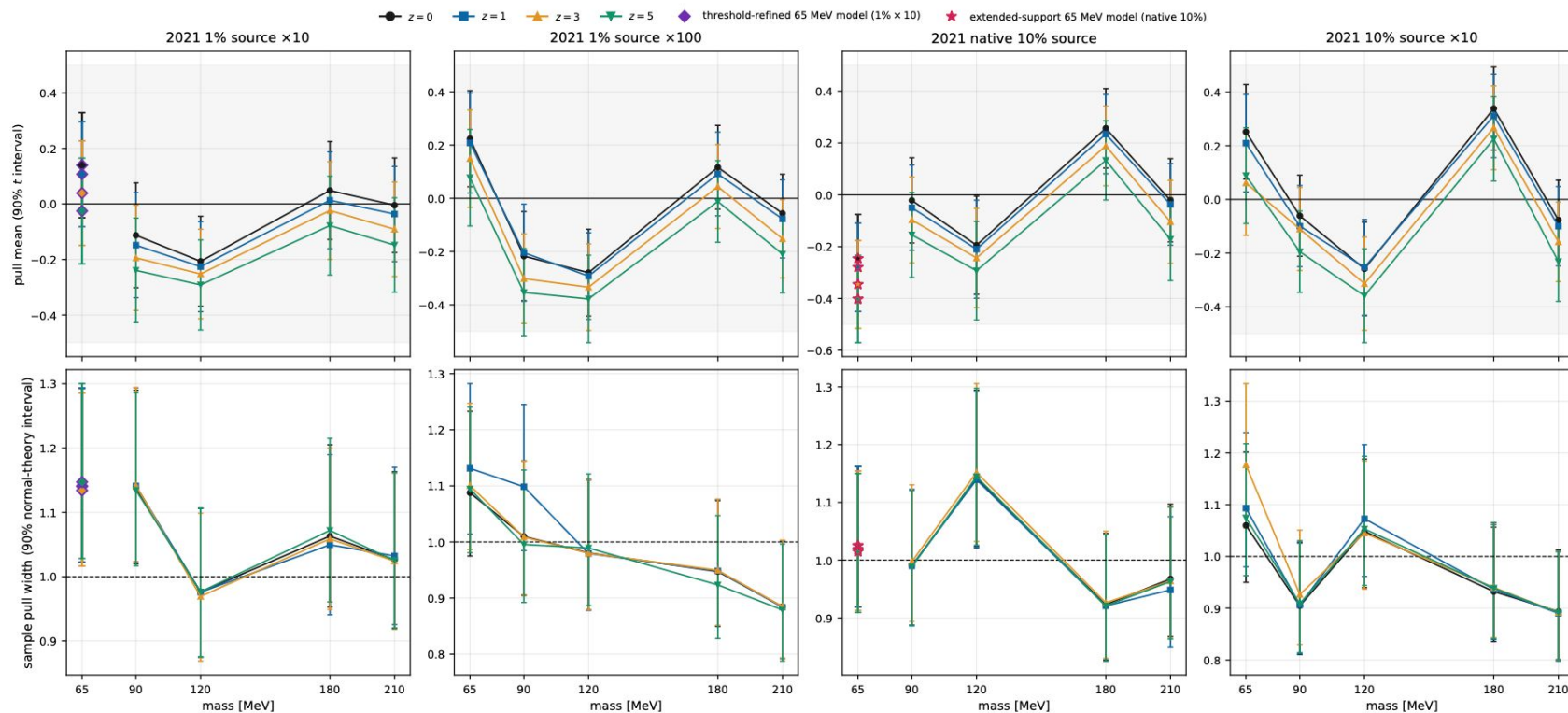
Explanation of threshold modeling and alternate form

[explain the different colored 1%*10 study and 10% study, use one figure for each explanation on this slide, explanations should be a few bullet points]

[This slide should explain exactly what is being done in signal injection and sigma reference and maybe a demonstrative display on that and briefly the refitting procedure]

Injected Signal Studies

Signal-injection pull diagnostics across 100 pseudoexperiments per point



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Bounded profile-likelihood ratio (non-negative signal)

$$q_{0,d}(m) = -2 \ln \frac{\mathcal{L}_d\left(A_d = 0, \hat{\hat{\theta}}_d(0)\right)}{\mathcal{L}_d\left(\hat{A}_d, \hat{\theta}_d\right)}$$

[one more statement about this equation
would be nice]

Observed Upper Limits

[briefly walk through this process in a few bullet points, looking through my previous presentations could help but make sure to make it easy to understand and maybe show a plot below that shows the upper limit point based on when the model crosses the CLs value of 0.1]

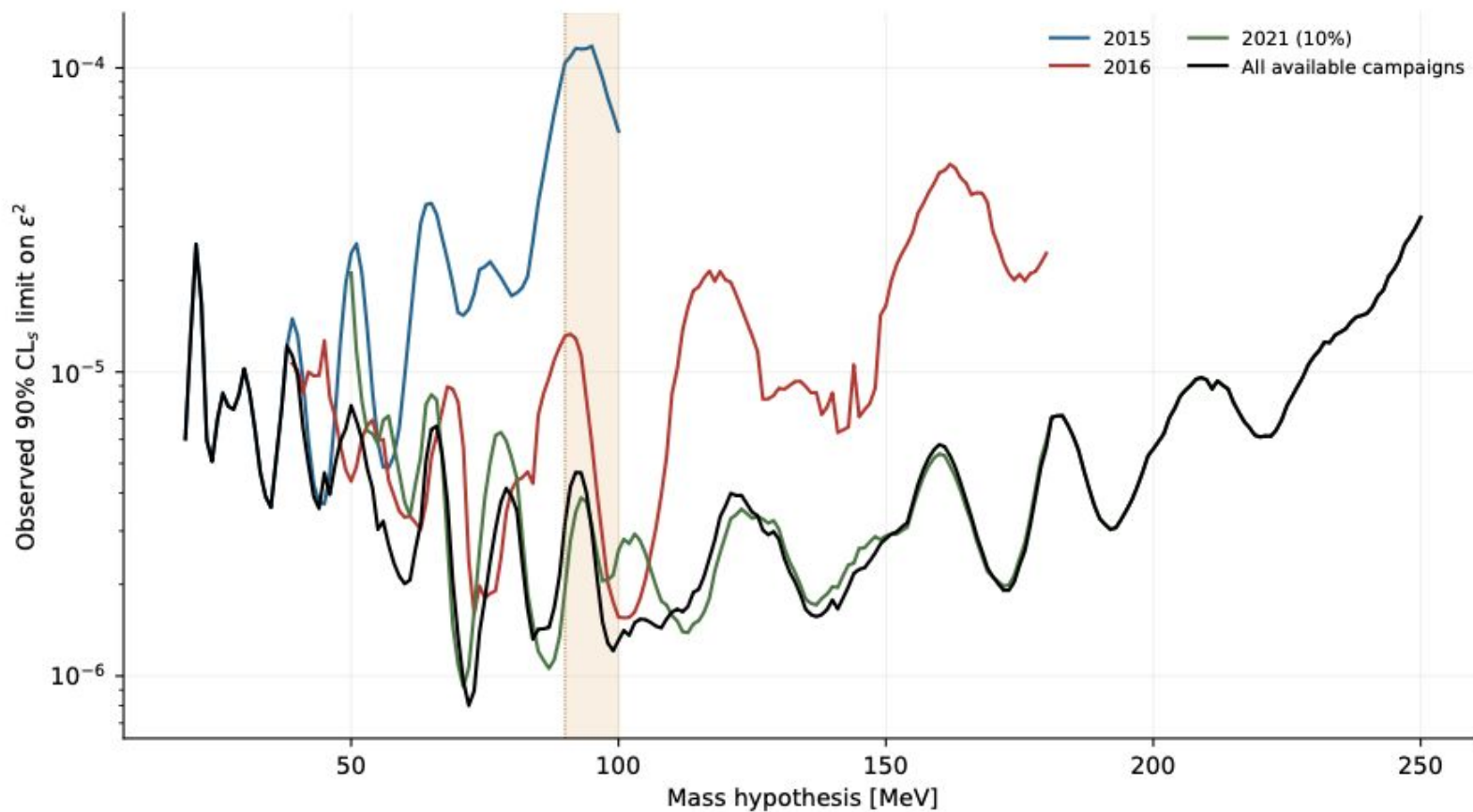
$$\tilde{q}_{A_d}(m) = \begin{cases} 0, & \hat{A}_d^{\text{unc}} > A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(\hat{A}_d^{\text{unc}}, \hat{\theta}_d)}, & 0 \leq \hat{A}_d^{\text{unc}} \leq A_d, \\ -2 \ln \frac{\mathcal{L}_d(A_d, \hat{\theta}_d(A_d))}{\mathcal{L}_d(0, \hat{\theta}_d(0))}, & \hat{A}_d^{\text{unc}} < 0, \end{cases}$$

$$\text{CL}_b(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | \text{bkg})$$

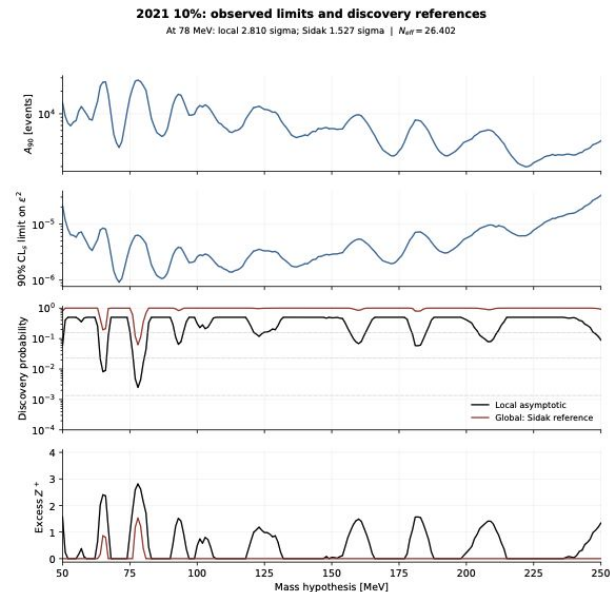
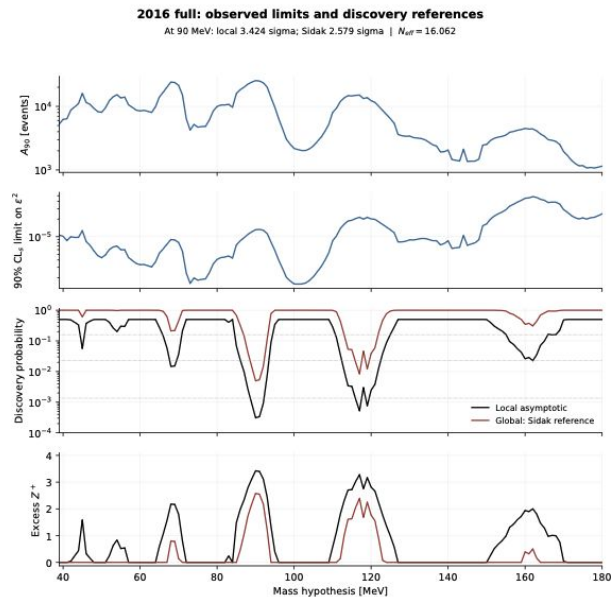
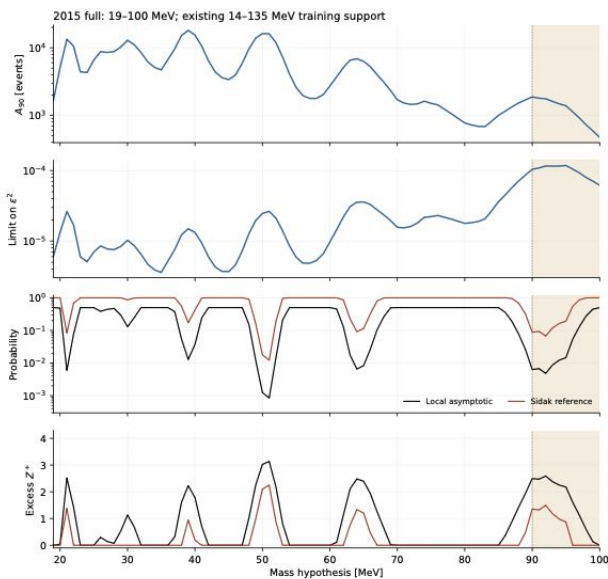
$$\text{CL}_{s+b}(A_d; m) = P(\tilde{q}_{A_d} \geq \tilde{q}_{A_d}^{\text{obs}} | A_d + \text{bkg})$$

$$\text{CL}_s(A_d; m) = \frac{\text{CL}_{s+b}(A_d; m)}{\text{CL}_b(A_d; m)}$$

Observed Upper Limits



Observed Limits and Discovery Pvalues for individual datasets

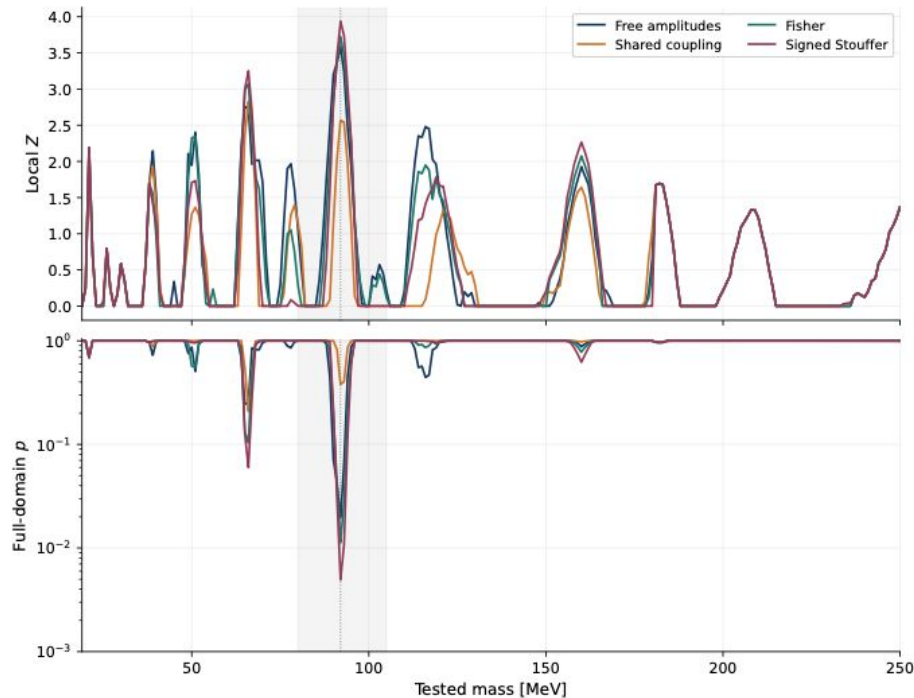


Combined discovery significance

Combined Significance dependent
on underlying phenomenological
framework

Shared Coupling: 2.8 sigma

Individual Experiments: 3.9 sigma

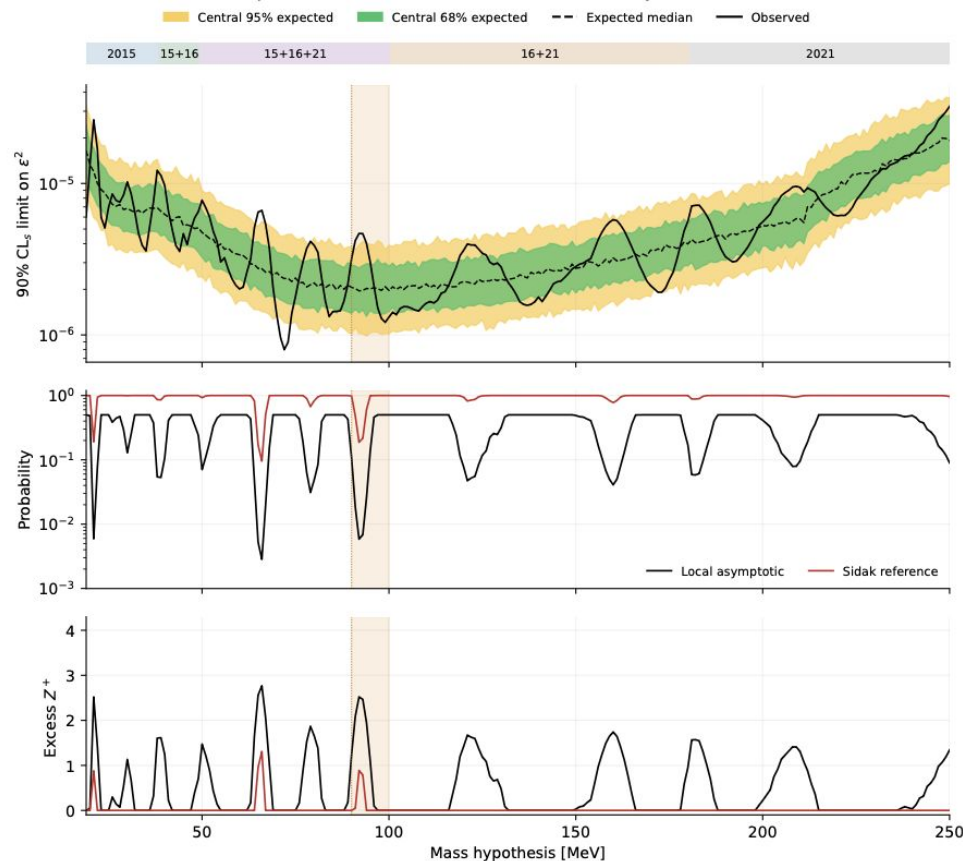


Combined Search Results

300 Toys for bands

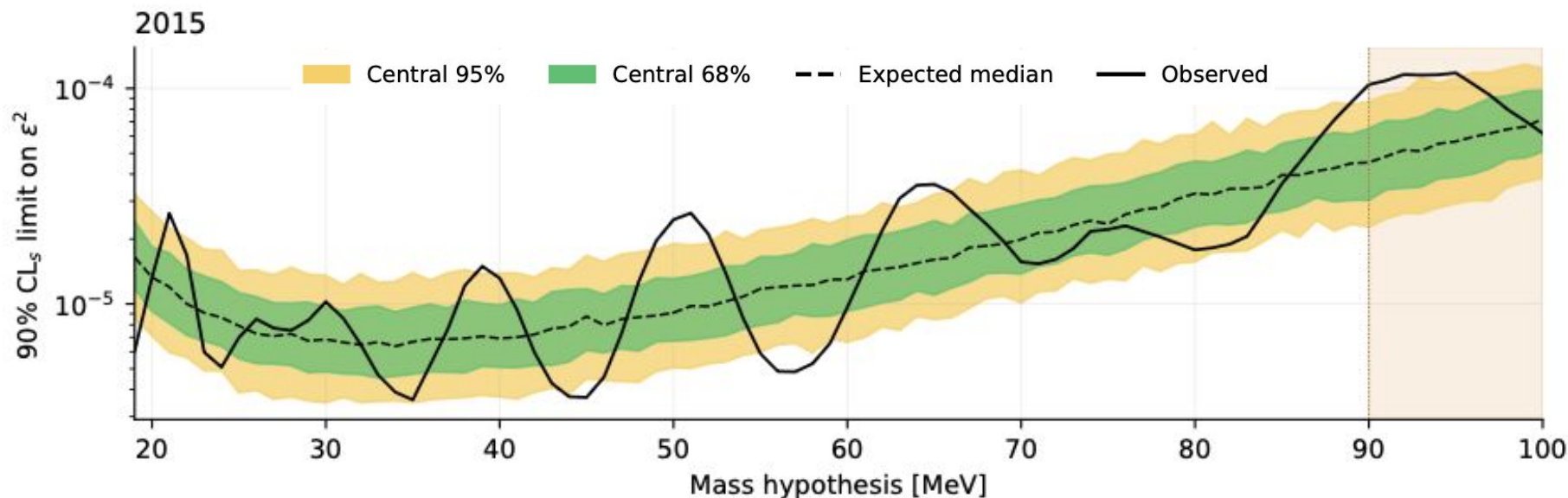
LEE penalty factor of 35

- Most significant local 2.8
- Most significant local 1.4

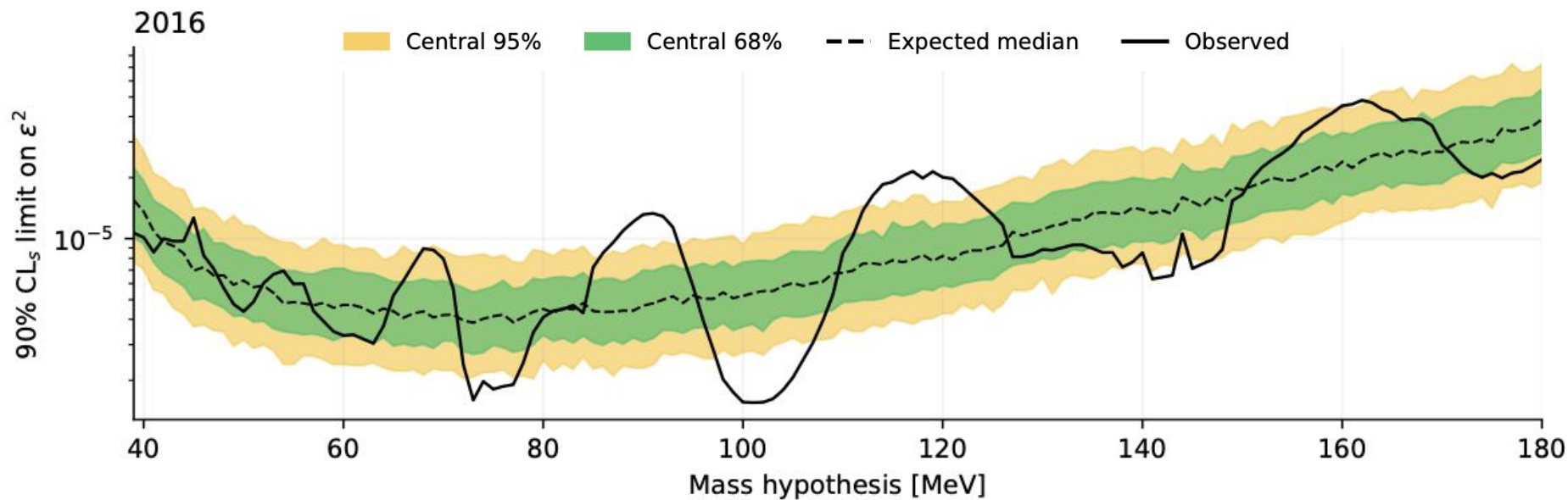


2015 Individual Limits

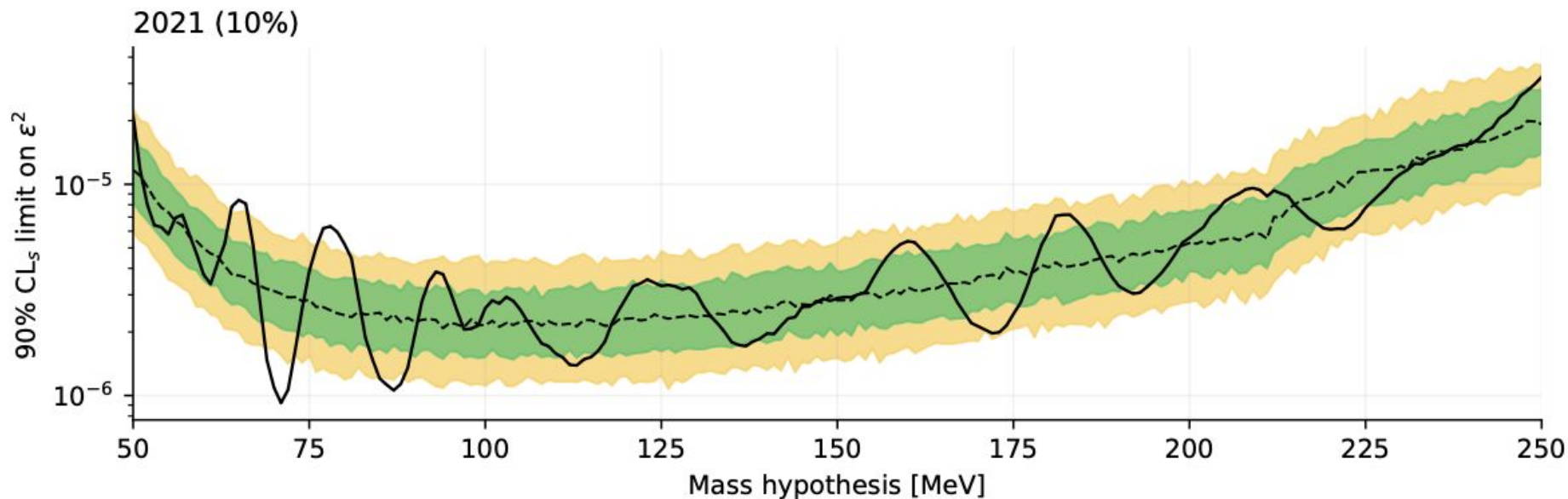
Comparison to re-analysis in backup



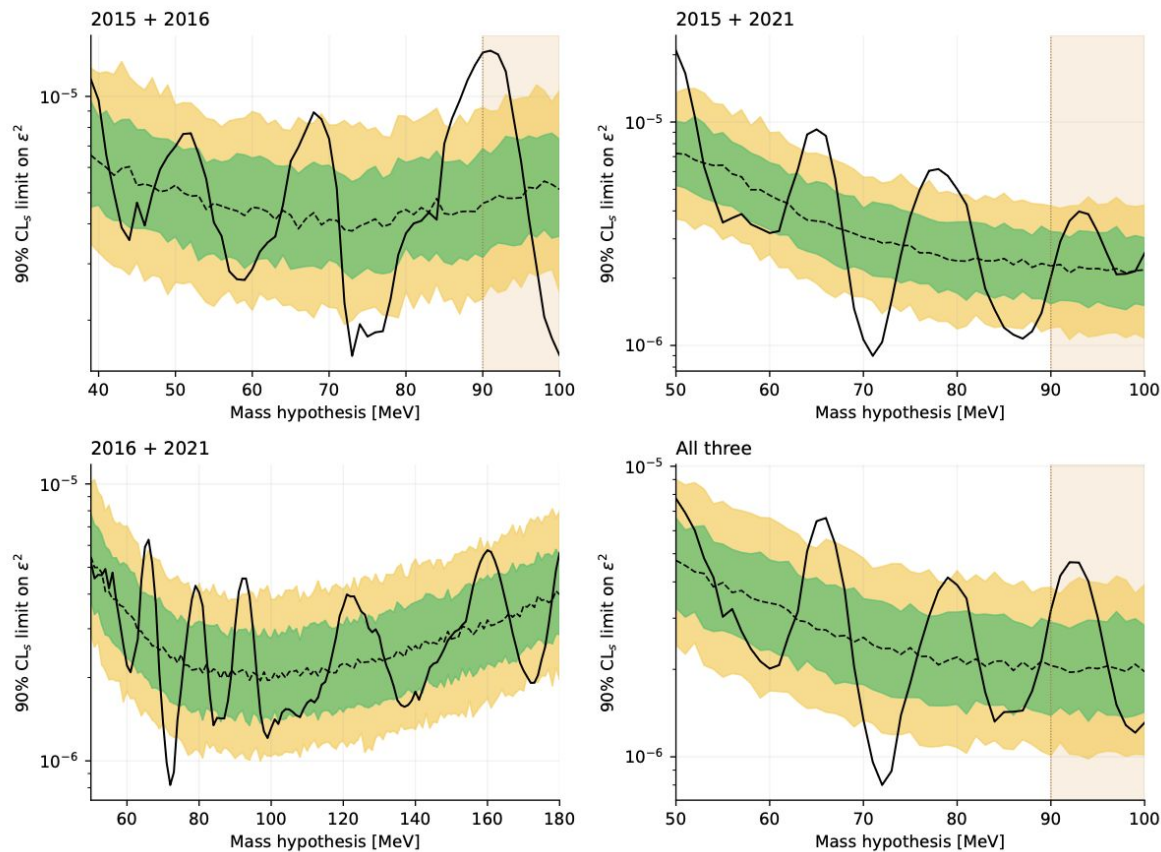
2016 Individual Limits



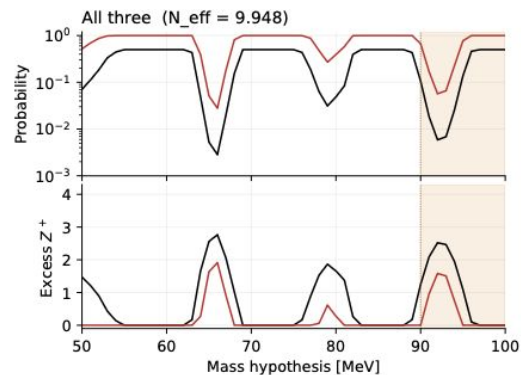
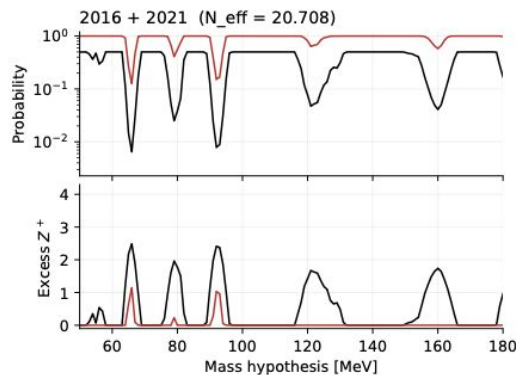
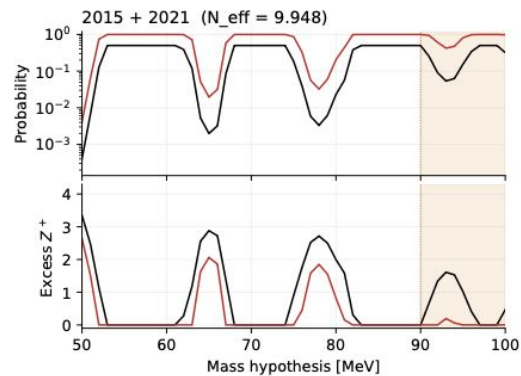
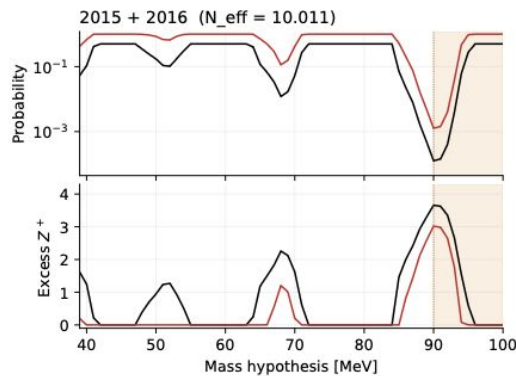
2021 10%



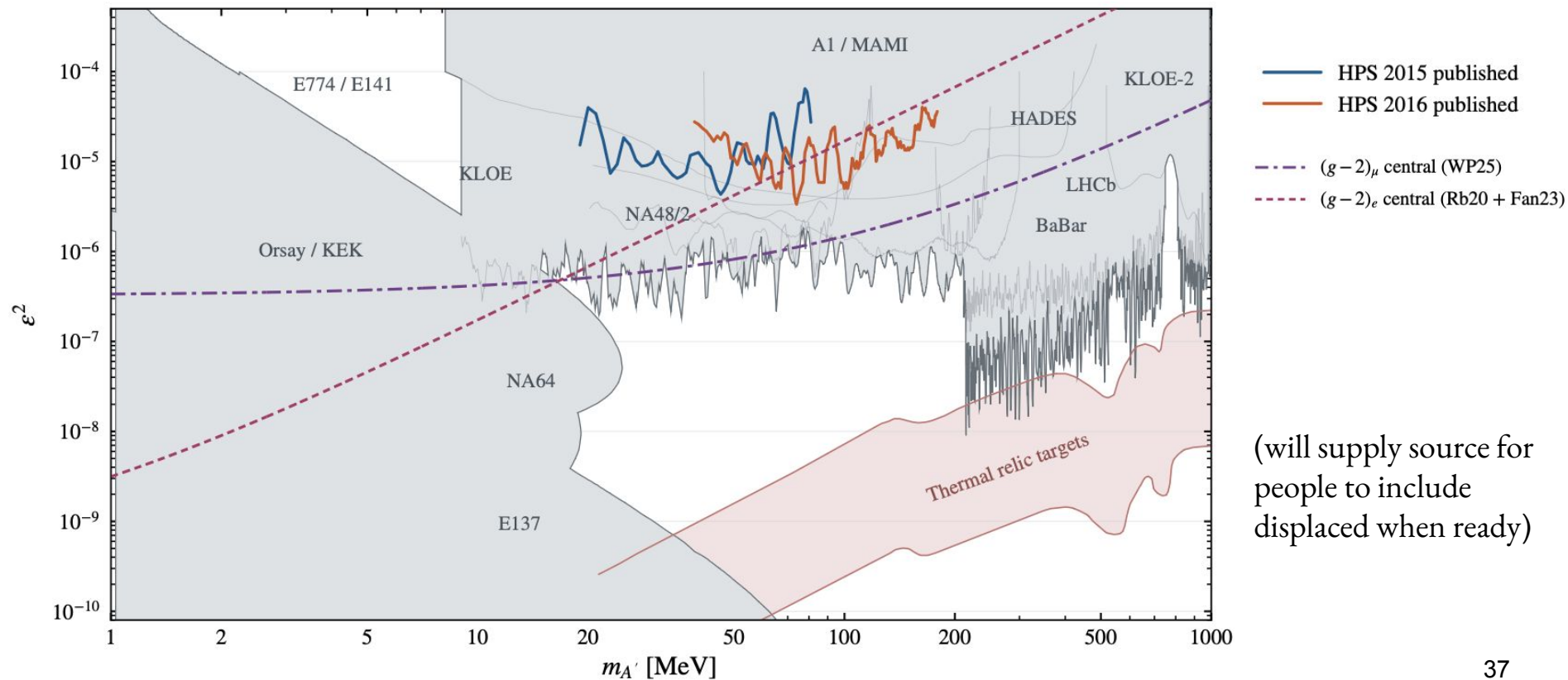
All Combined Regions



Combined region discovery pvalues

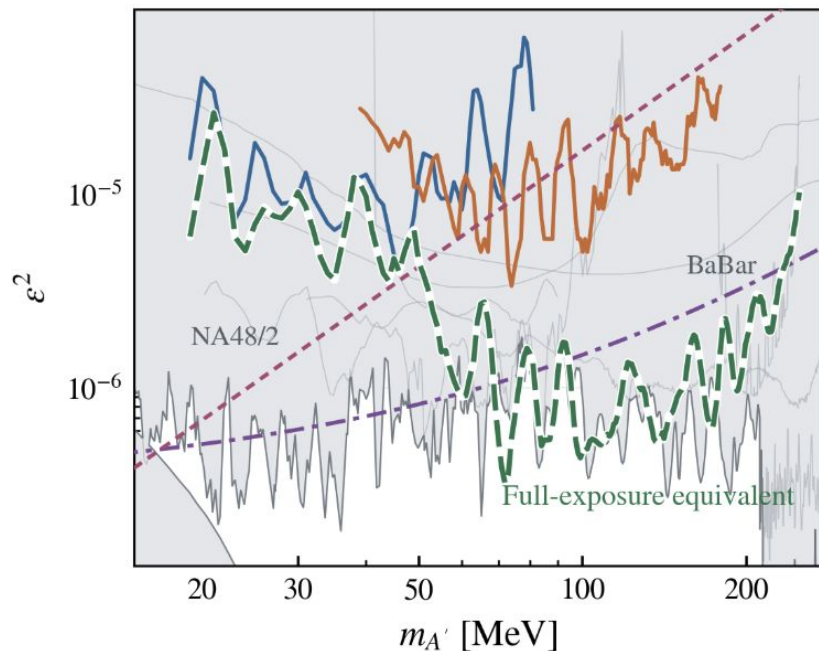


Making Cleaner and Modern Contours

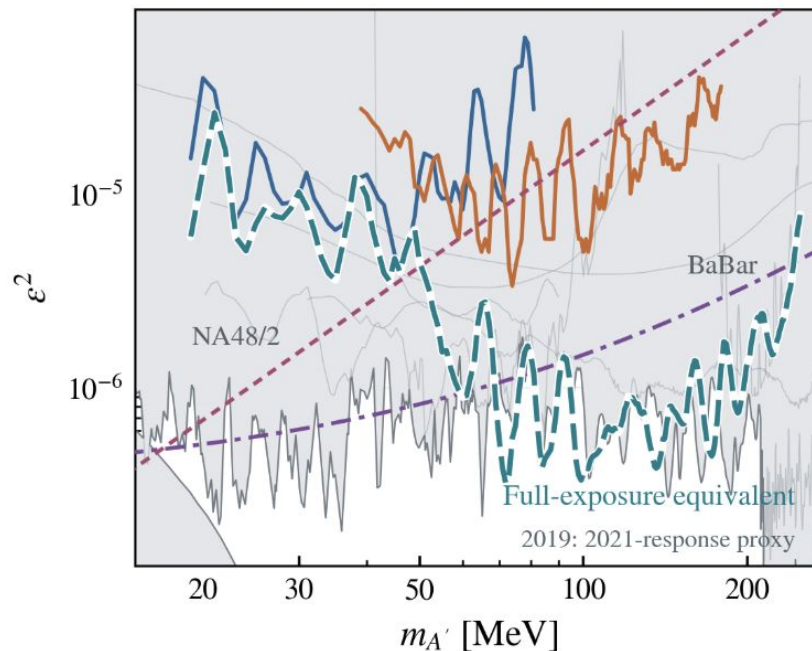


Projected Contours

(a) 2015 + 2016 + 2021



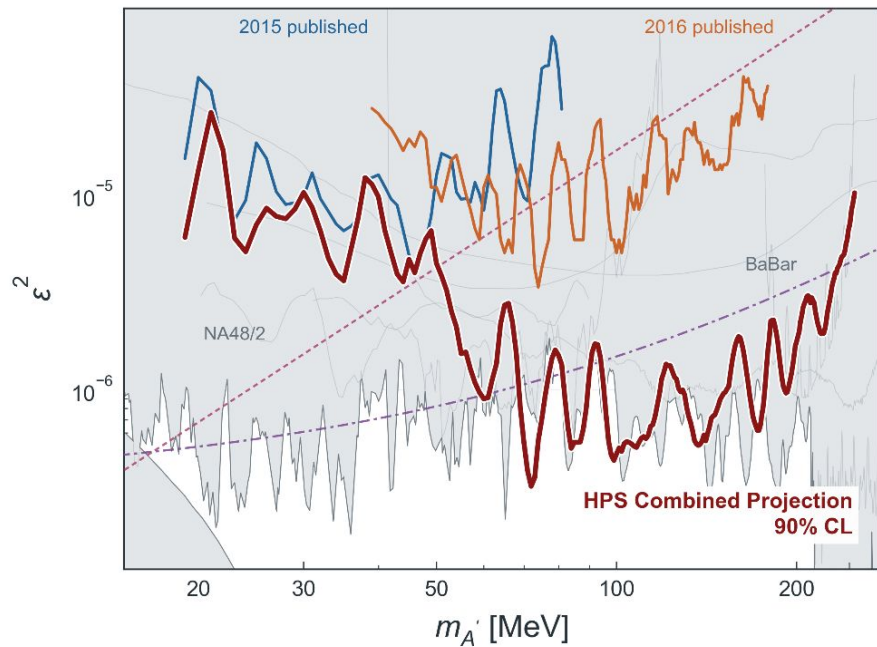
(b) 2015 + 2016 + 2019 + 2021



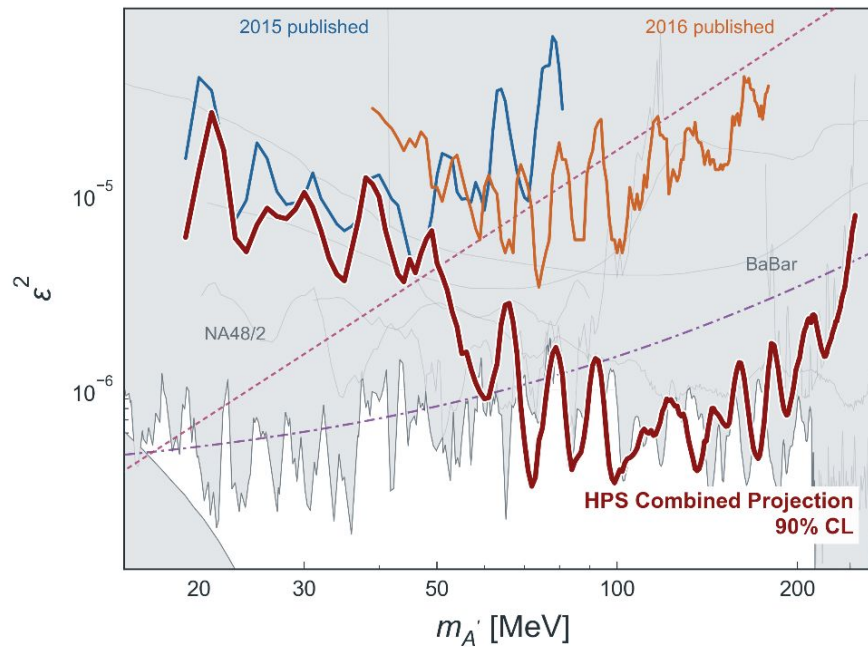
More thorough babar comparison at end of presentation.

2019 comes from 1% UC sample, used for luminosity equivalent and ~coverage only.

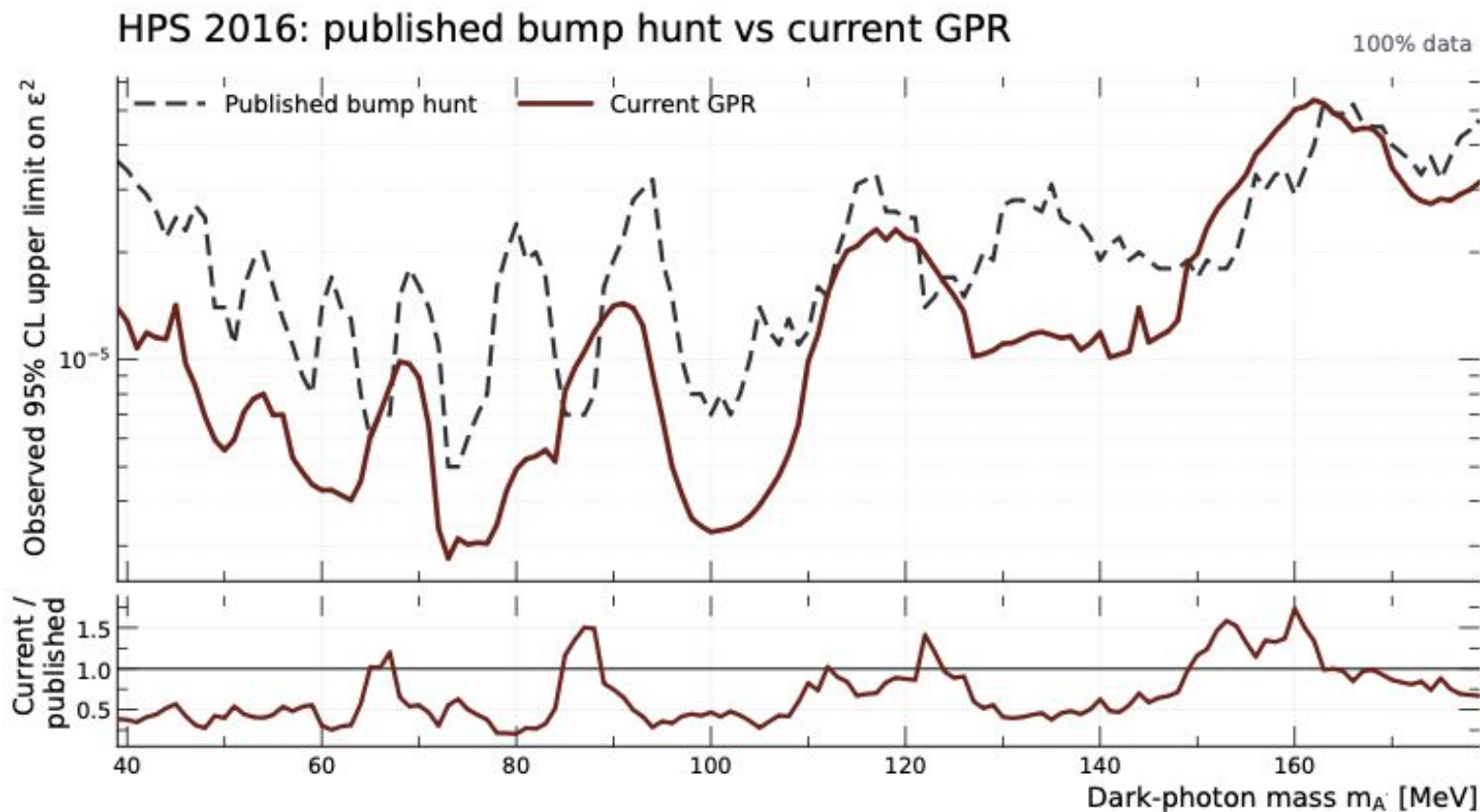
(a) 2015 + 2016 + 2021



(b) 2015 + 2016 + 2019 + 2021



Observed Limit Comparison with 2016 Publication



What would a signal look like in full 2021 dataset?

- must be able to map correlations if they exist in invariant mass
- must be able to characterize shape of observed upper limit with signal injected

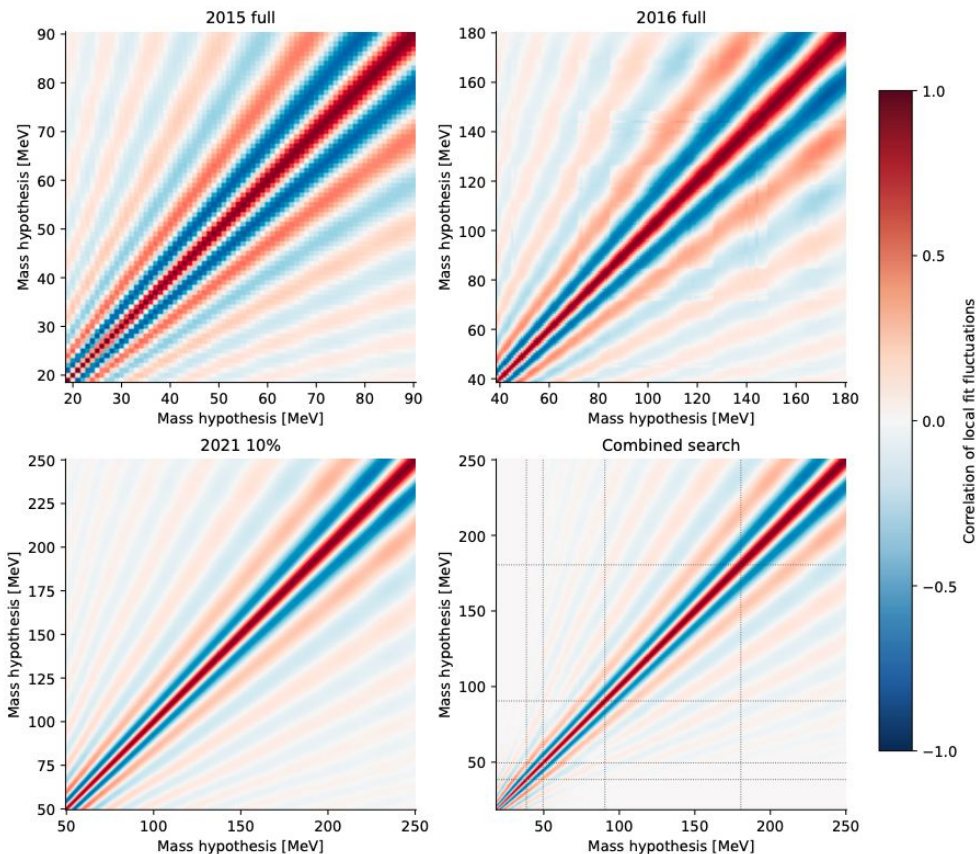
What is the max significance a signal could be in 2021?

GPR Correlations in mass ($1/n$)

This mapping shows that for a local 1 sigma fluctuation there is a strong anti-correlation nearby that oscillates as you move away

Makes sense!

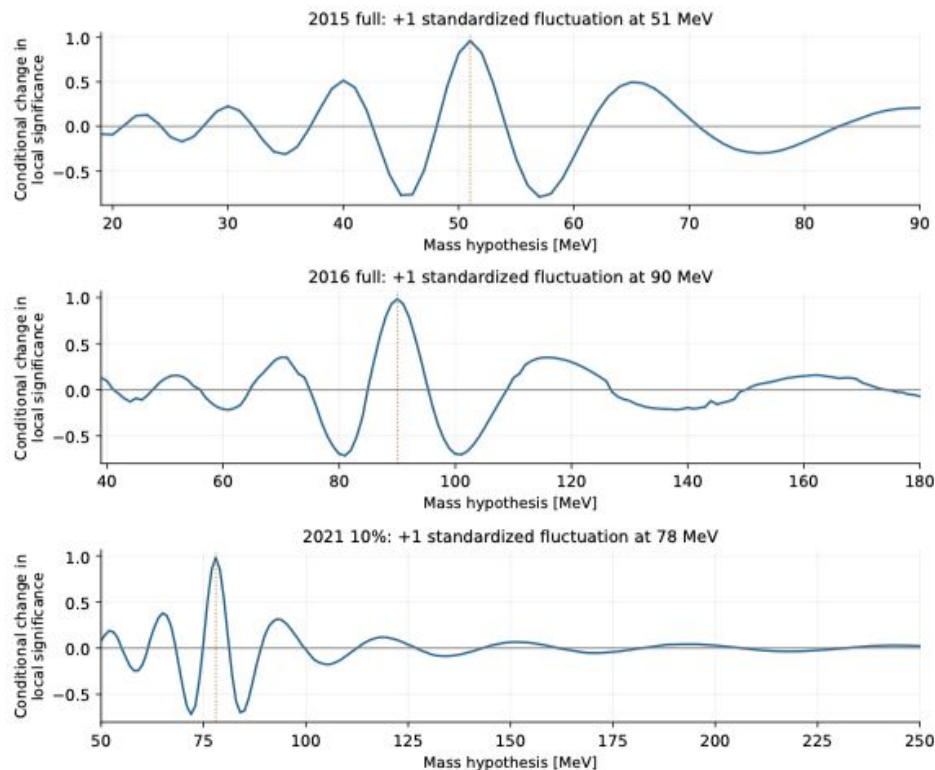
Signal in a training sideband will of course spoil the training nearby



GPR Correlations in mass ($1/n$)

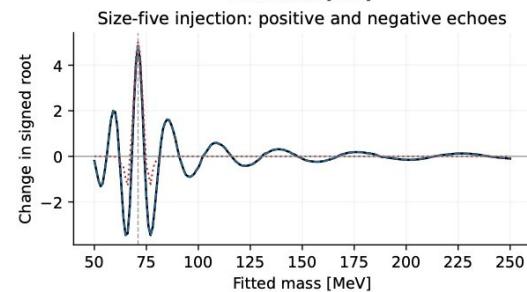
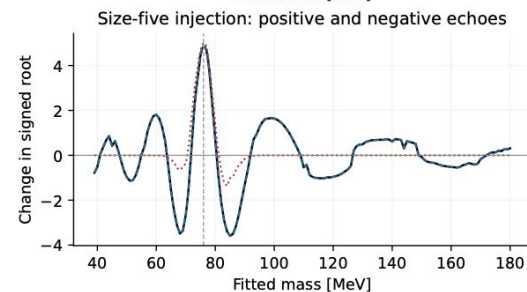
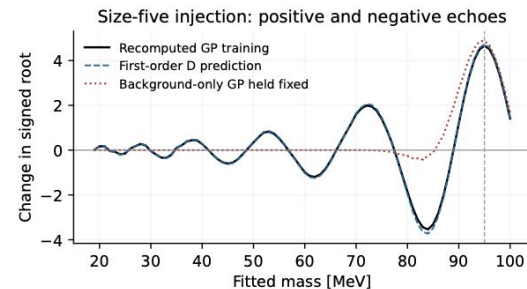
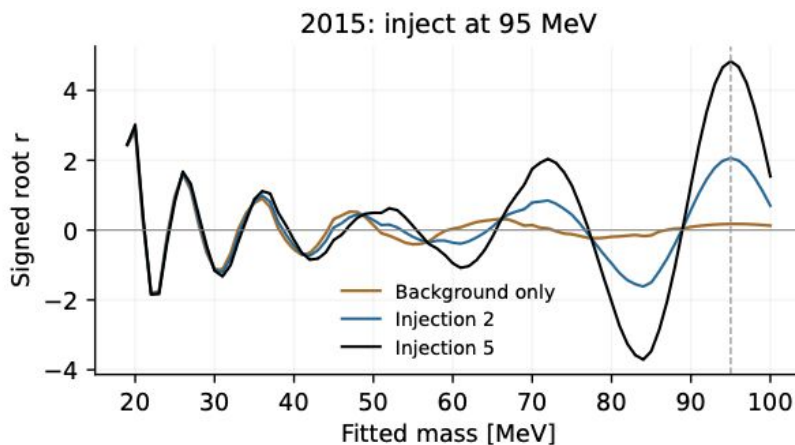
Size/shape of correlated
“echoes” dependent on

- mass resolution
- location of peak
- underlying background of nearby regions



First order prediction for location of features

$$\Delta r_m^{\text{lin}} = \sum_{i: B_i > 0} D_{im} \frac{\delta n_i}{\sqrt{B_i}}$$



Projection of max signal if regions of interest are true signal

$$S_{100\%} = kS_f, \quad B_{100\%} = kB_f, \quad Z_{100\%} \simeq \frac{kS_f}{\sqrt{kB_f}} = \sqrt{k} Z_f$$

$$Z_{1\% \rightarrow 100\%}^{\text{target}} = 10Z_{1\%}$$

$$Z_{10\% \rightarrow 100\%}^{\text{target}} = \sqrt{10} Z_{10\%}.$$

Predicting Echoes for Unblinded 2021

Its possible to first order [here please explain what teh below screenshot is doing]

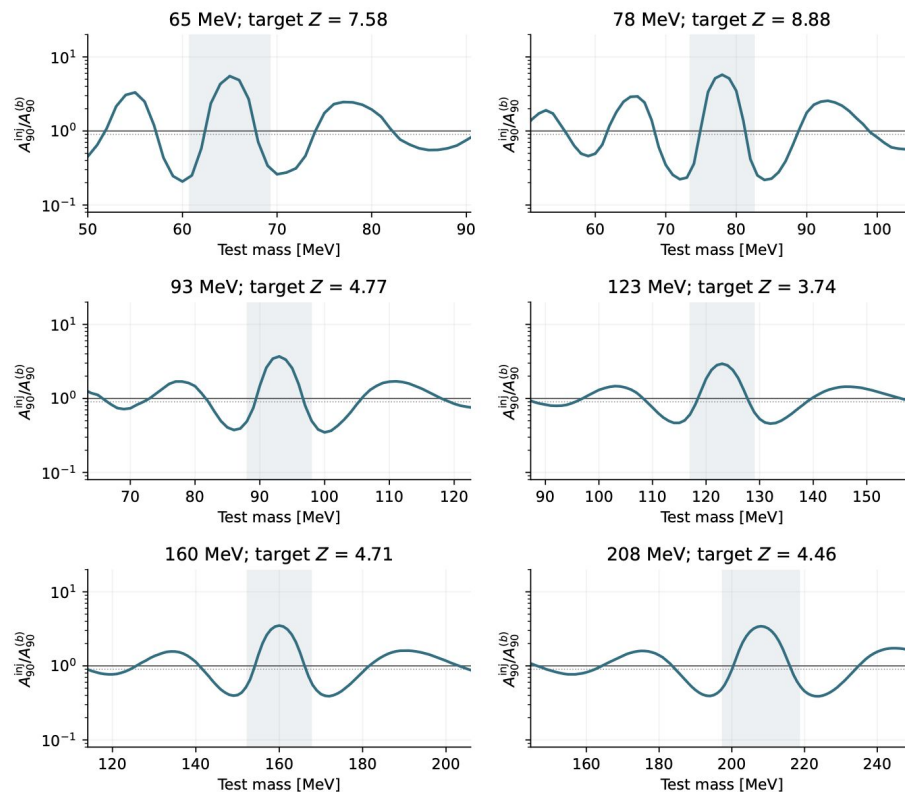
- GP fit to 10% distribution, threw 20 toys with same statistics as 100%
- add signal template on top scaled to expected significance [referencing previous slide]
- poisson draw ($b_i + s_i$)

$$R(m) = \frac{A_{90}^{\text{injected}}(m)}{A_{90}^{(b)}(m)}, \quad D = 1 - R(m_{\text{echo}}), \quad m_{\text{echo}} \in m_0 \pm [2, 8] \sigma_m(m_0)$$

Predictions of upper limit structure

Have proven ability to predict
observed upper limit structure to
first order

projected shape of upper limits for
different regions of interest in full
2021 dataset



- I. Additional Local Likelihood using GP on projected significance
- II. 2021 Threshold fit optimization
- III. Signal Extraction Display
- IV. Mass Resolution Studies
- V. 2015 95% CLs Comparison
- VI. Projection to Babar Plot
- VII. Two Signal Hypothesis
- VIII. Potential fit catalogues
- IX. Interpretation slides

Local Likelihood using GP on projected significance: pt1

- New approach to throw toys that make sense of correlations in gpr [cite here and show few equations motivating use]

[show plots that work through steps of determining local and global significance]

Local Likelihood using GP on projected significance: pt2



- New approach to throw toys that make sense of correlations in gpr [cite here and show few equations motivating use]

[show plots that work through steps of determining local and global significance]

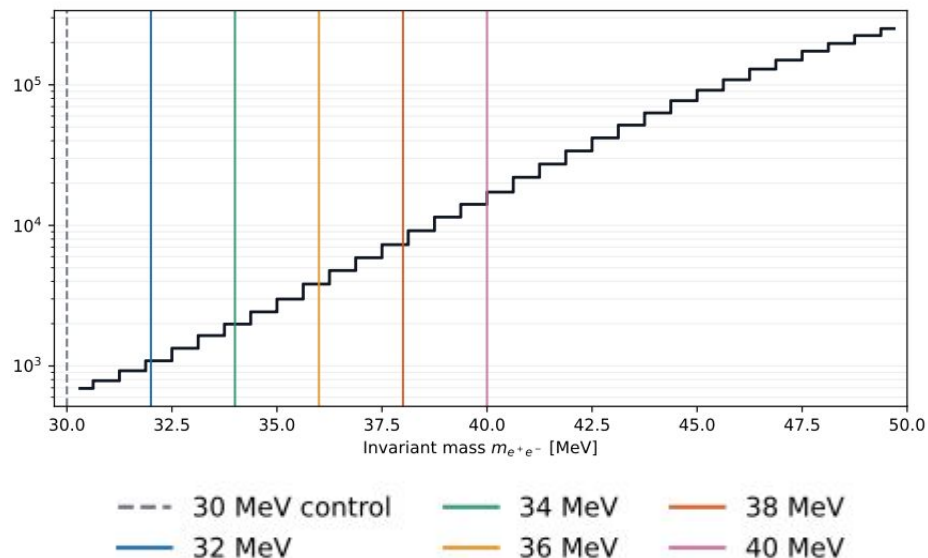
2021 Threshold Fit optimization

Study done after seeing apparent biases with GP fit support starting at 40 MeV in pseudoexperiment tests.

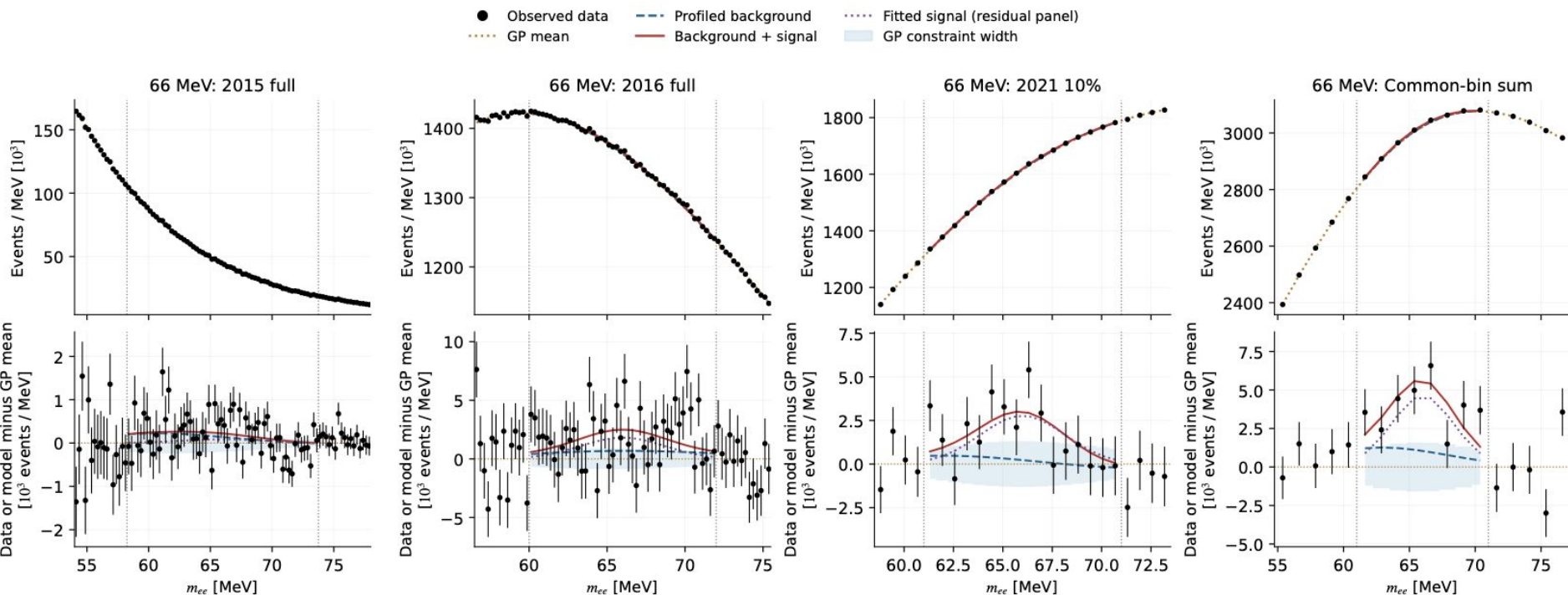
- selected GP fit support range and procedure for selection for fully unblinded 2021 dataset

Rising edge support selected: 36 MeV
Note goes into more detail

10% 2021 Threshold Region



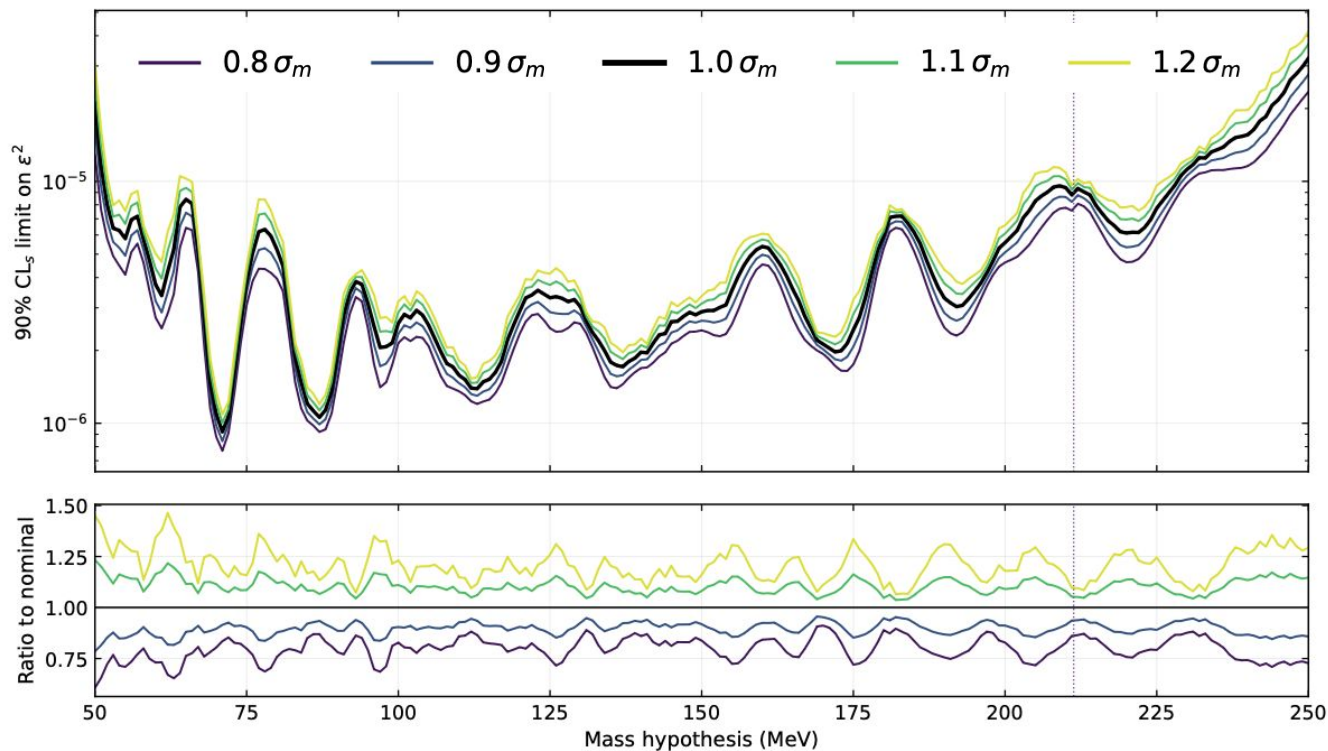
Extraction Display



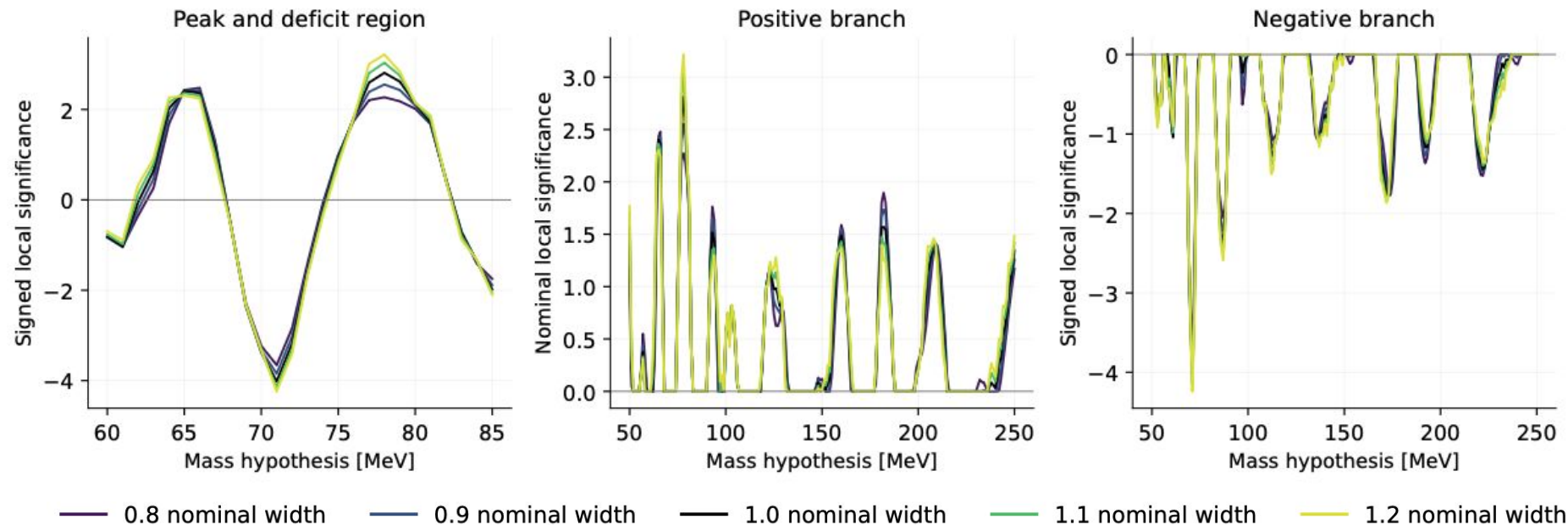
Wiggling the mass resolution pt 1

Impacts

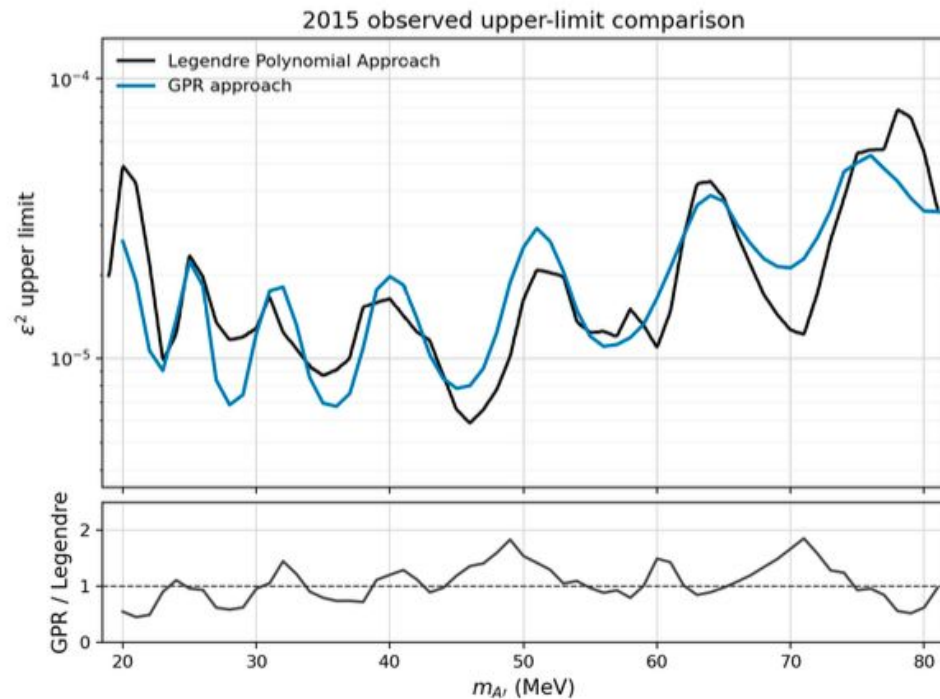
- blind window size
 - (fit uncertainty)
- signal width
- bkg density calculation (marginally)



Wiggling the mass resolution pt 2



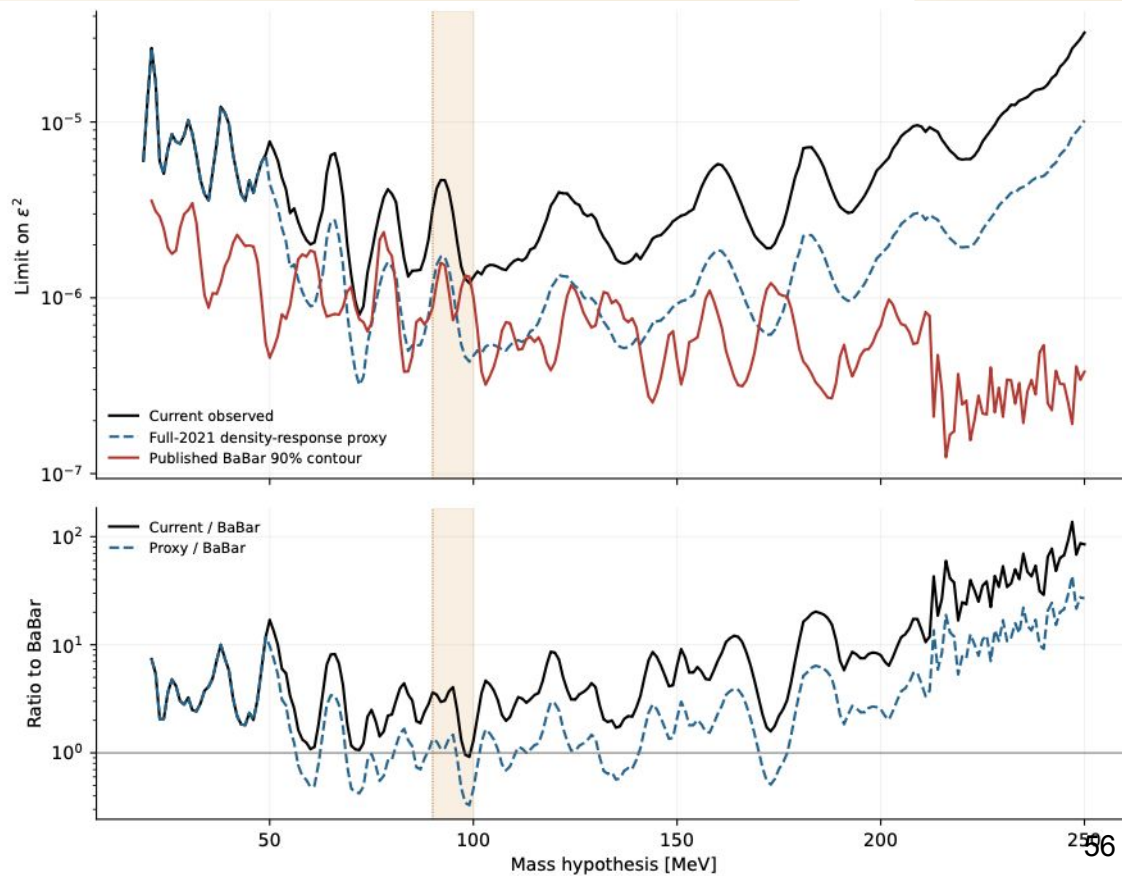
2015 Comparison



Babar Comparison

Currently have small
region of new exclusion

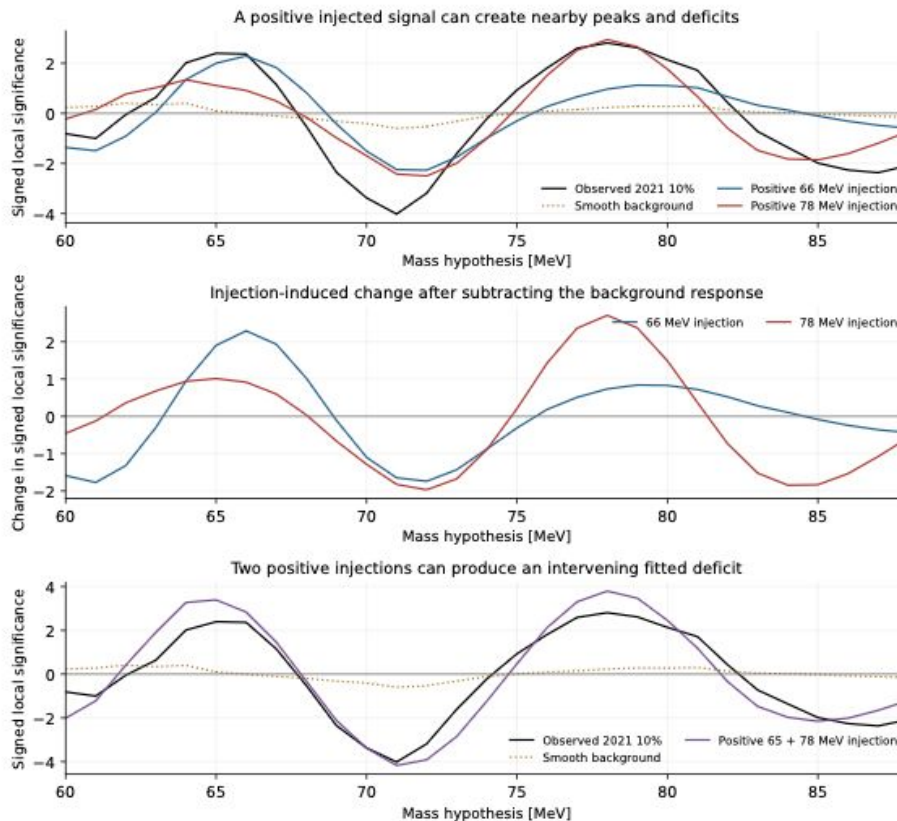
- Exciting!



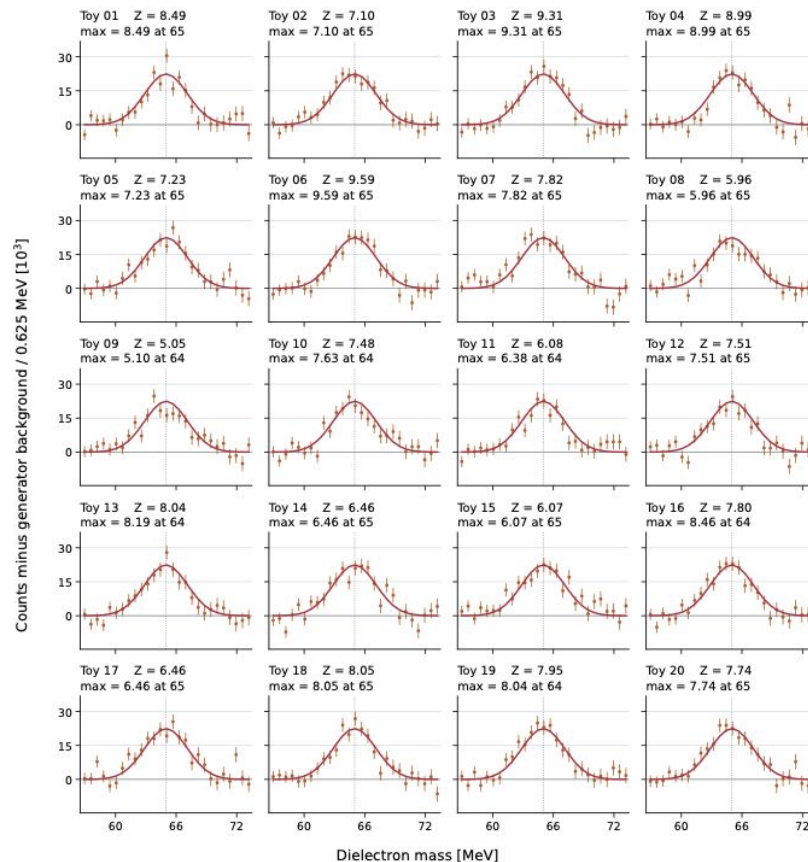
Two Signal Modeling

Two signals can create constructive/destructive effects in observed upper limit

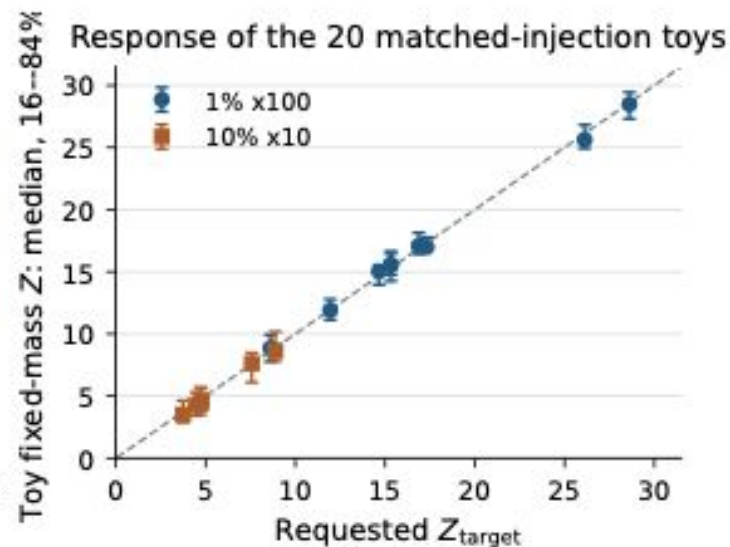
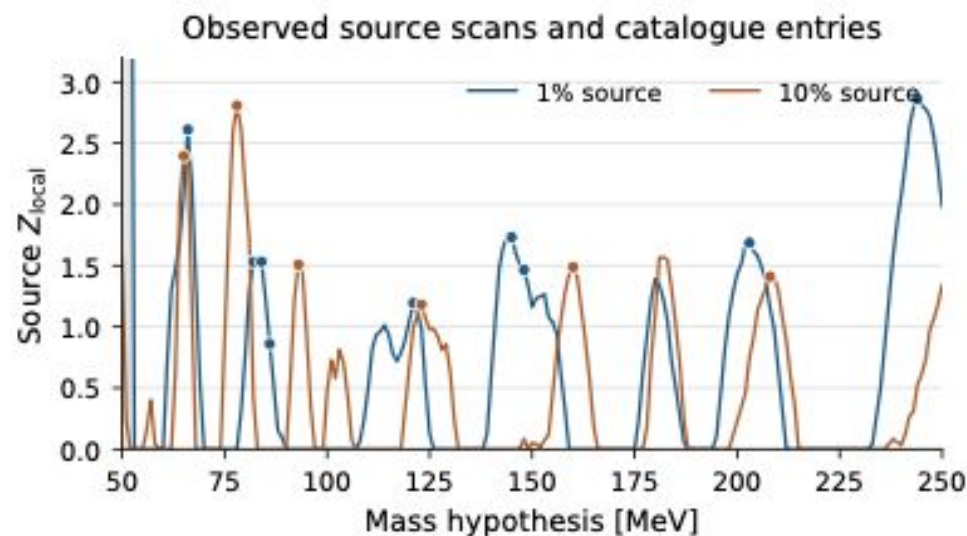
- 2 mild excesses could create deeper deficit than a single nearby signal



hypothetical fit catalogue (65 MeV toys)



Catalogue overview



All “Candidates”

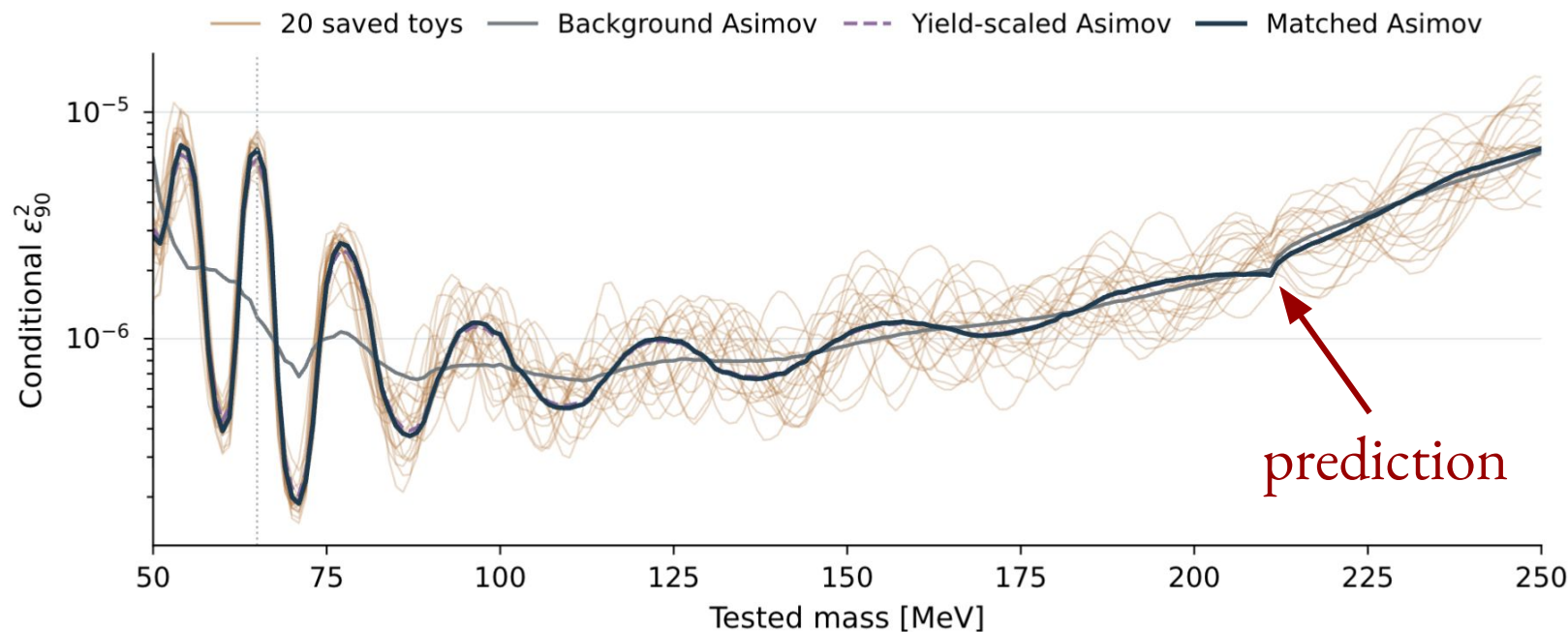
Source	Region [MeV]	m_0 [MeV]	Z_f	Target	Z_A^{yield}	R_a	Toy median [16,84%] fixed mass	Max ₂₀ fixed mass
1%	65	66	2.61	26.12	23.43	1.12	25.62 [24.93, 26.82]	27.79
10%	65	65	2.40	7.58	6.76	1.12	7.63 [6.10, 8.47]	9.59
1%	75	82	1.53	15.32	13.99	1.10	15.54 [14.25, 16.47]	17.92
10%	75	78	2.81	8.88	8.39	1.06	8.64 [7.80, 10.18]	11.62
1%	92	86 [†]	0.86	8.61	7.99	1.08	8.86 [7.77, 9.92]	10.45
10%	92	93	1.51	4.77	4.45	1.07	4.63 [3.92, 5.58]	6.33
1%	120	121	1.19	11.94	11.13	1.07	11.91 [11.13, 12.79]	13.04
10%	120	123	1.18	3.74	3.64	1.03	3.51 [2.89, 4.63]	4.92
1%	160	148 [†]	1.47	14.66	13.59	1.08	15.03 [13.96, 15.53]	16.53
10%	160	160	1.49	4.71	4.54	1.04	4.39 [3.50, 5.71]	7.35
1%	210	203	1.69	16.87	15.50	1.09	17.09 [16.43, 18.13]	19.02
10%	210	208	1.41	4.46	4.16	1.07	4.23 [3.41, 5.30]	6.90
1%	+84	84	1.53	15.33	14.05	1.09	15.57 [14.75, 16.65]	17.05
1%	+145	145	1.73	17.32	16.07	1.08	16.98 [16.78, 17.76]	18.77
1%	+244	244	2.86	28.61	25.31	1.13	28.48 [27.31, 29.45]	30.30

Predictions of observed upper limit

following catalogue shows that we are able to predict shapes of upper limits to first order!

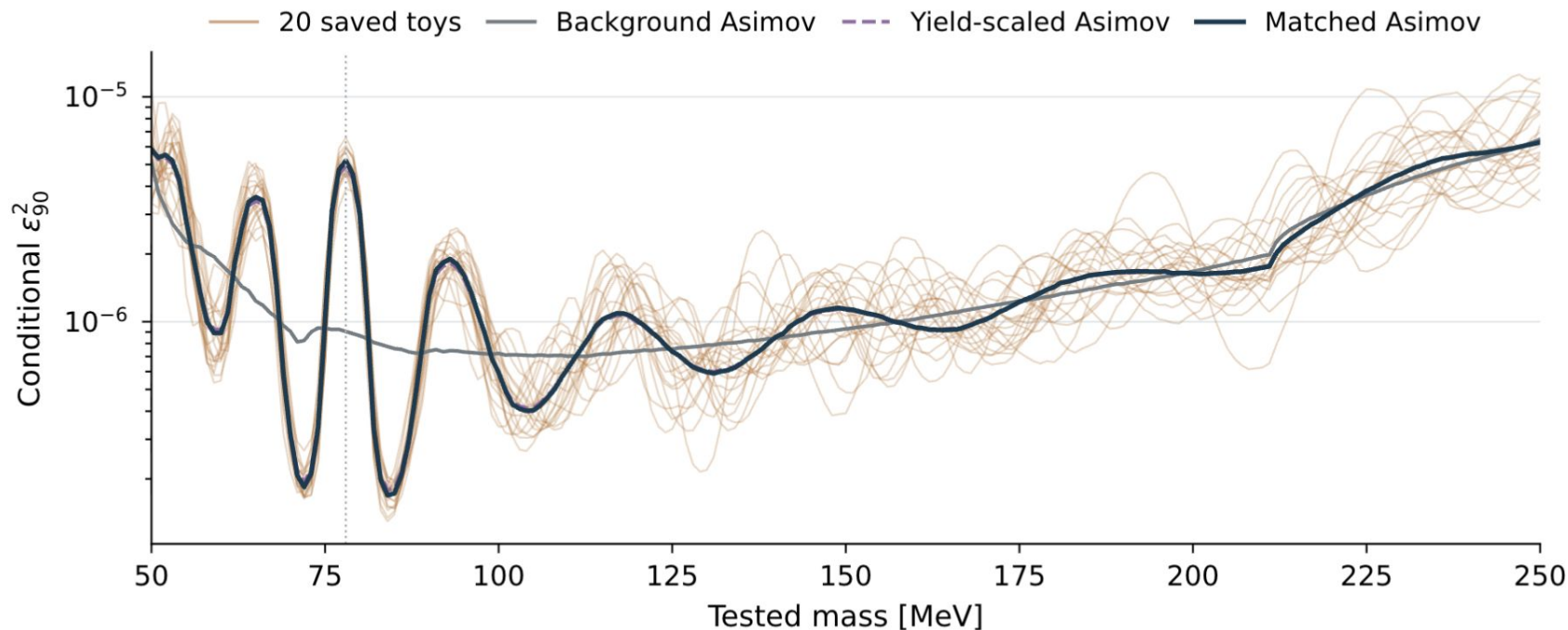
100% Prediction: 65 MeV

Median Projected Local Significance: 7.6 Sigma



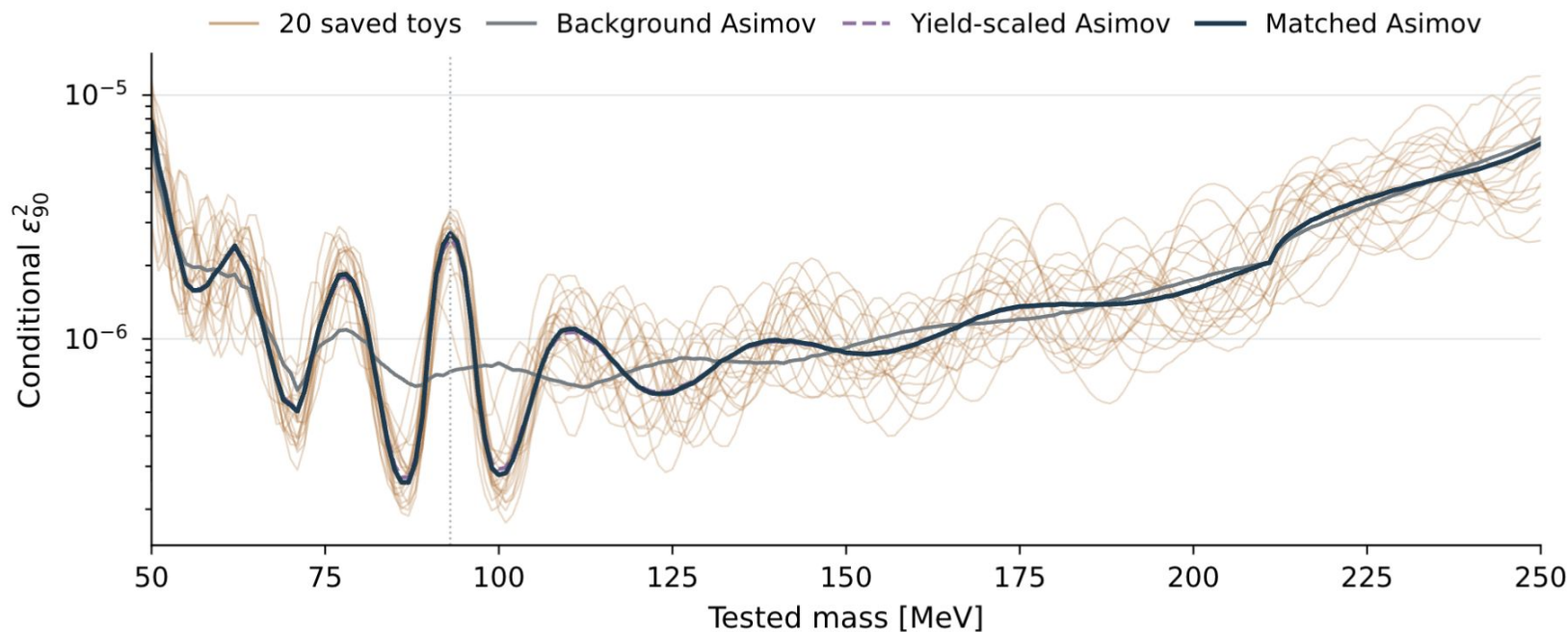
100% Prediction: 78 MeV

Median Projected Local Significance: 8.88 Sigma



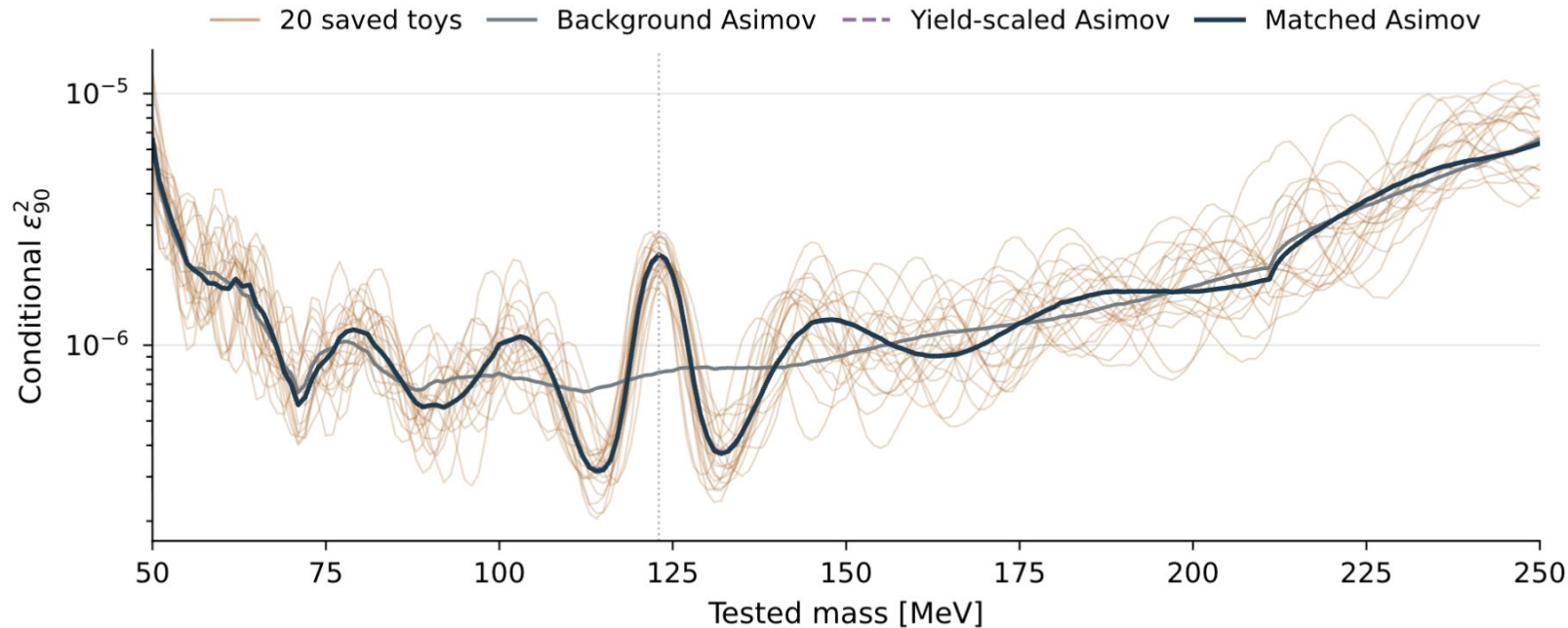
100% Prediction: 93 MeV

Median Projected Local Significance: 4.77 Sigma



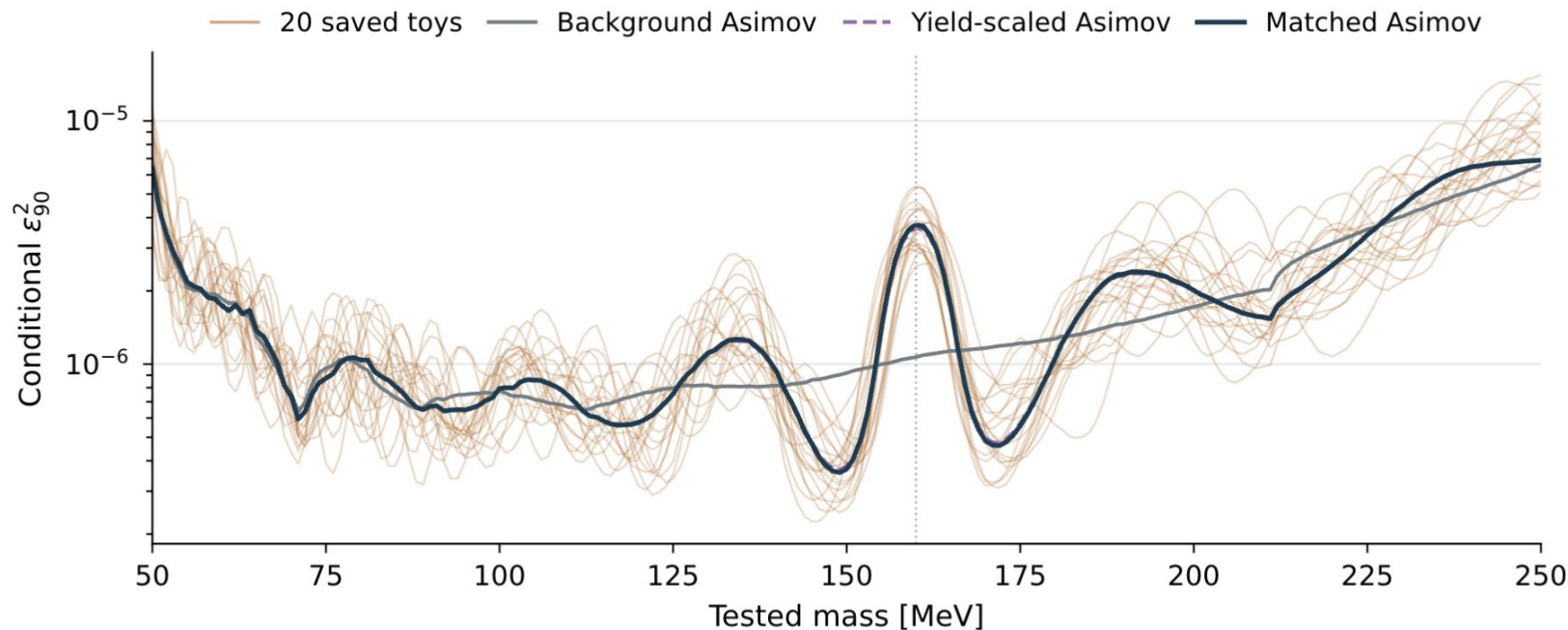
100% Prediction: 120 MeV

Median Projected Local Significance: 3.74 Sigma



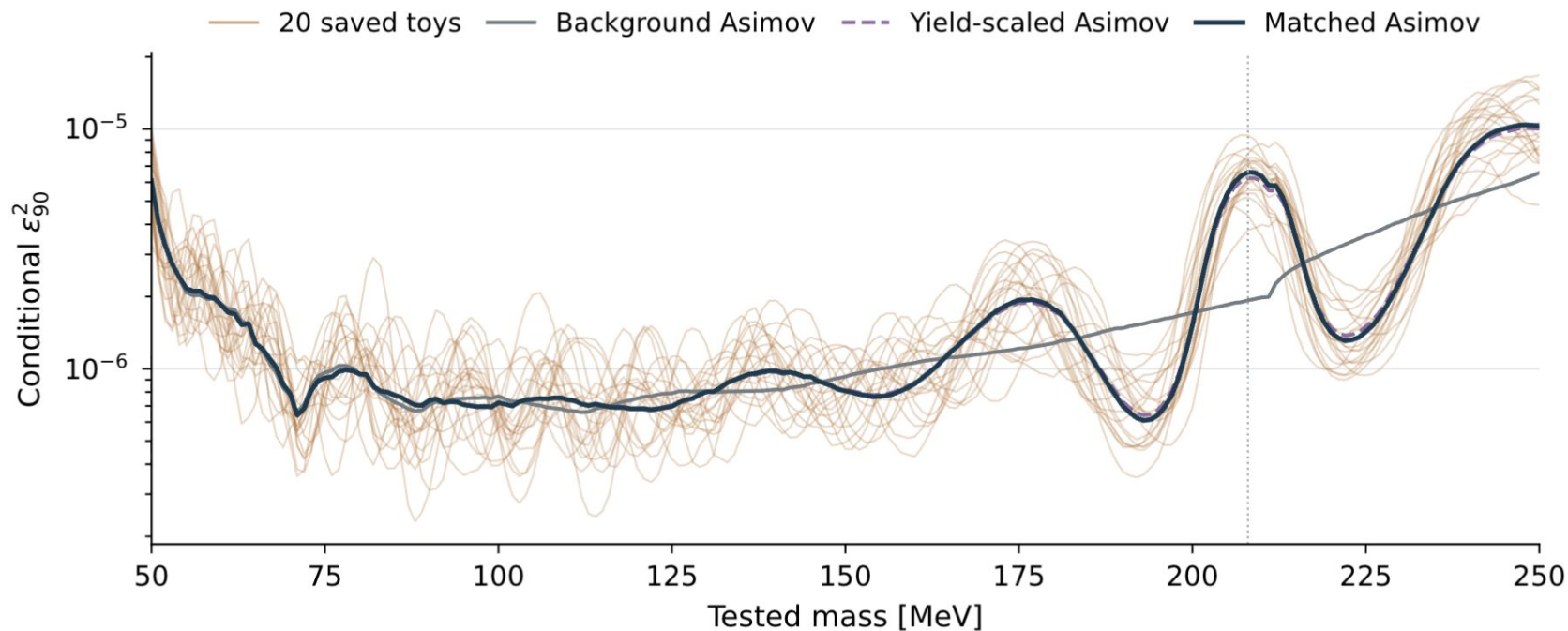
100% Prediction: 160 MeV

Median Projected Local Significance: 4.71 Sigma



100% Prediction: 208 MeV

Median Projected Local Significance: 4.46 Sigma



Interpretations

Loss metric

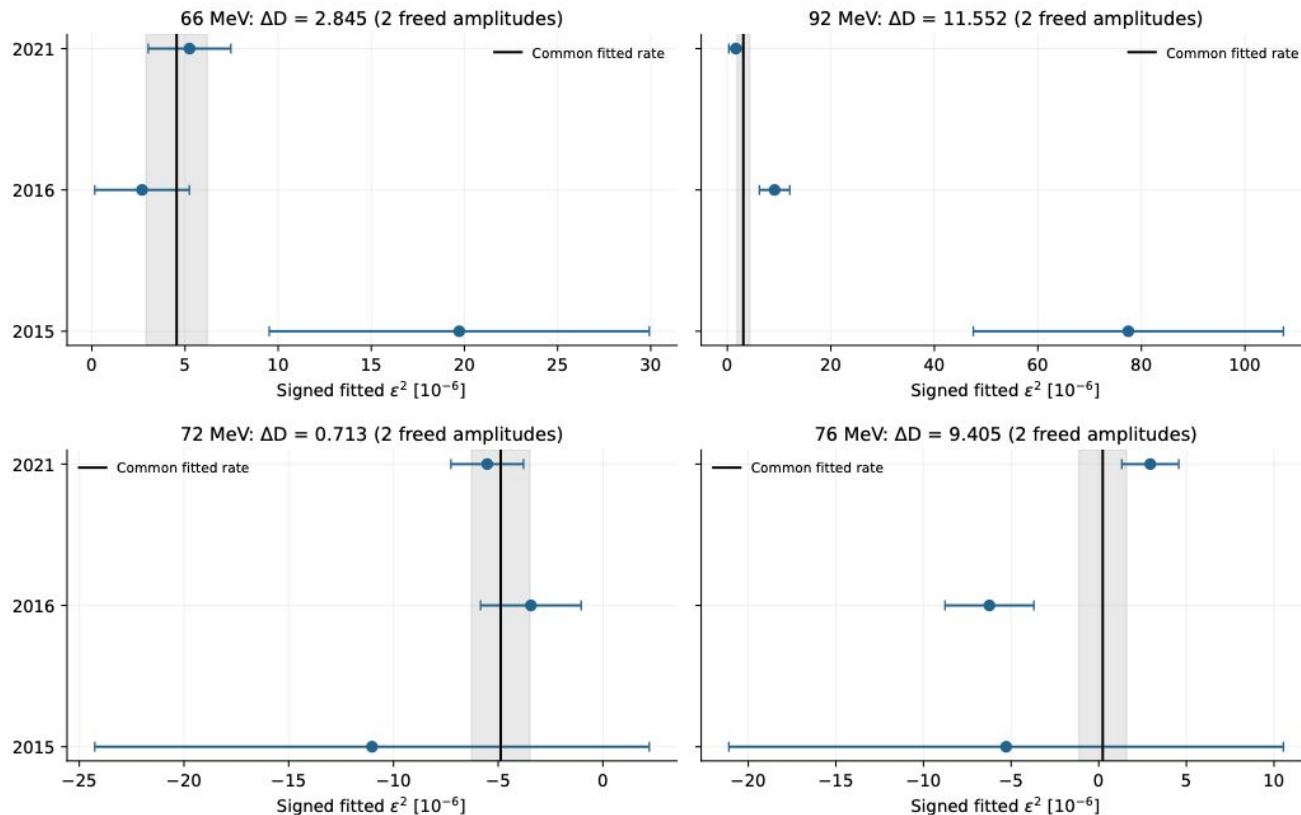
- large loss implies individual datasets describe observation better than shared coupling

$$\Delta D = 2 \left[\text{NLL}_{\text{common}} - \sum_d \text{NLL}_{d,\text{independent}} \right]$$

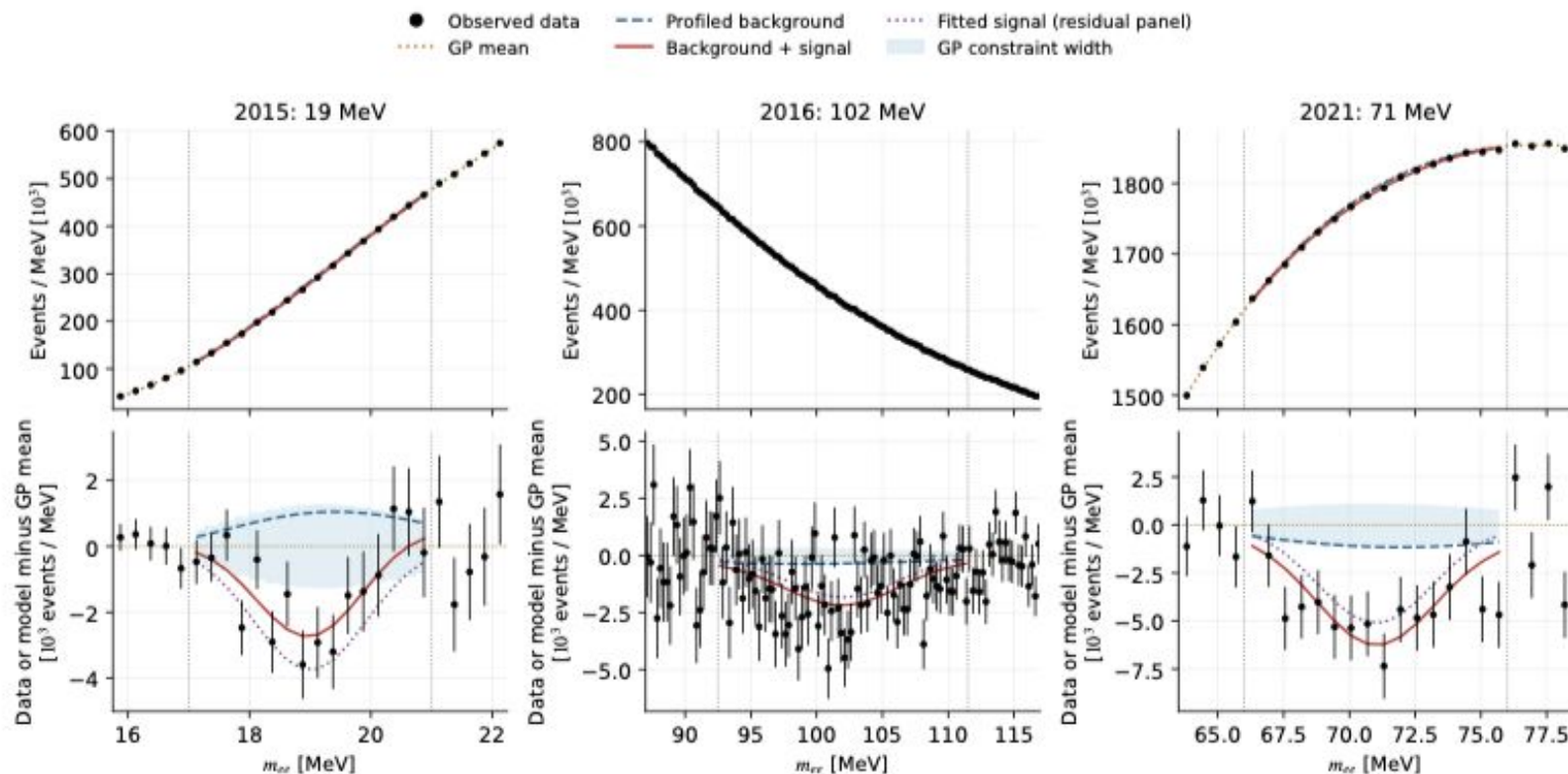
Common rate before 2015 Extension



Common rate after 2015 extension



Leading Deficits



Signal Model for alternative frameworks

Interpolated bkg

Beam Energy

$$\lambda_i = b_i + L_i \theta_i + \mu K_i(m_i, \sigma_i) (E_i / 2.30 \text{ GeV})^\beta$$

power law

Radiative Fraction + dataset dependent info

Profiled Background

The diagram shows the equation $\lambda_i = b_i + L_i \theta_i + \mu K_i(m_i, \sigma_i) (E_i / 2.30 \text{ GeV})^\beta$ with five red lines pointing to specific parts: 'Interpolated bkg' points to b_i ; 'Beam Energy' points to E_i ; 'power law' points to the exponent β ; 'Radiative Fraction + dataset dependent info' points to $\mu K_i(m_i, \sigma_i)$; and 'Profiled Background' points to $L_i \theta_i$.

Things to consider post-unblinding

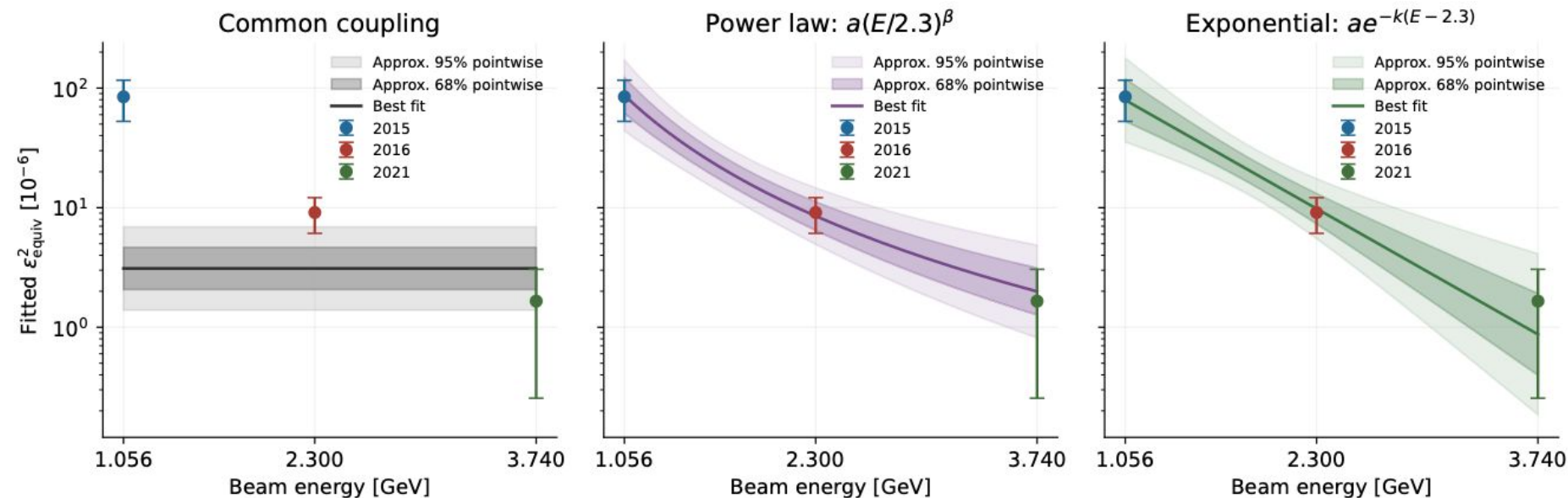
For a fixed common region of interest at 92 MeV

$$Q_G = \sum_i \max(z_i, 0)^2$$

Model or rule	$\sqrt{Q_G}$	Local p_G	Z_G
Common coupling	2.447	7.21×10^{-3}	2.447
Independent positive amplitudes	4.195	1.33×10^{-4}	3.646
Fitted power law	4.182	6.90×10^{-5}	3.812
Fitted exponential	4.148	7.77×10^{-5}	3.782
Equal-weight signed Stouffer	–	3.71×10^{-5}	3.962
Fisher of signed one-sided probabilities	–	6.91×10^{-5}	3.811

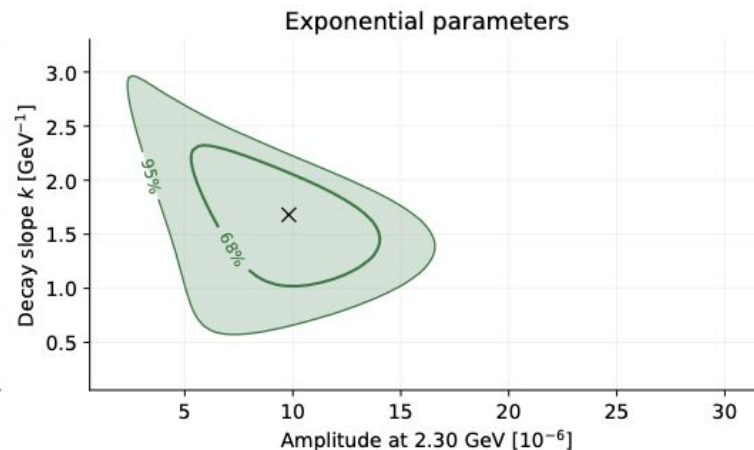
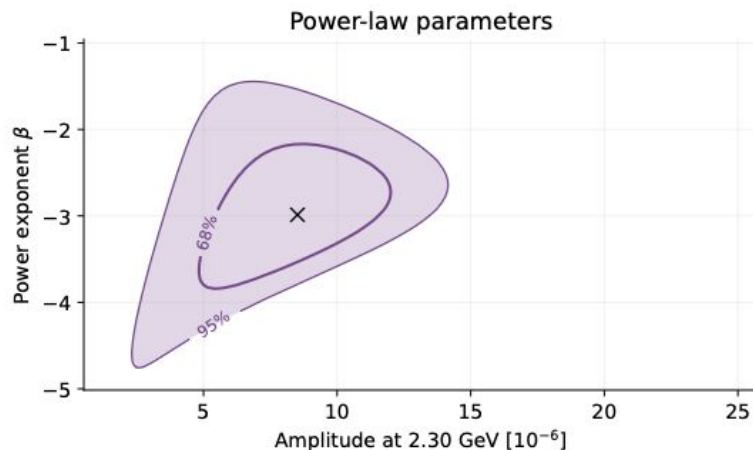
Alternative Combination Methods

Beam energy and Rate normalization assumptions

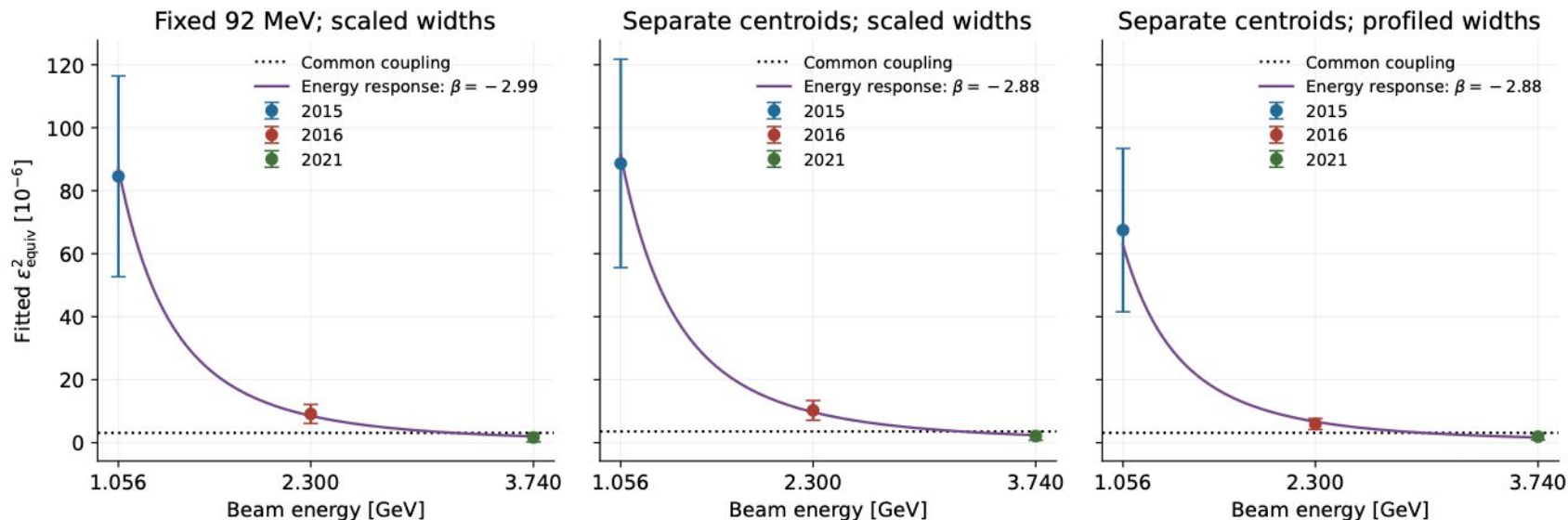


Two Parameterizations

$R(E)$	$\hat{a}$	conditional slope interval, 68% profile
1	3.102×10^{-6}	none
$(E/2.30)^\beta$	8.516×10^{-6}	$\beta = -2.987 [-3.517, -2.470]$
$\exp[-k(E - 2.30)]$	9.817×10^{-6}	$k = 1.681 [1.257, 2.087] \text{ GeV}^{-1}$



Profiling centroid and widths

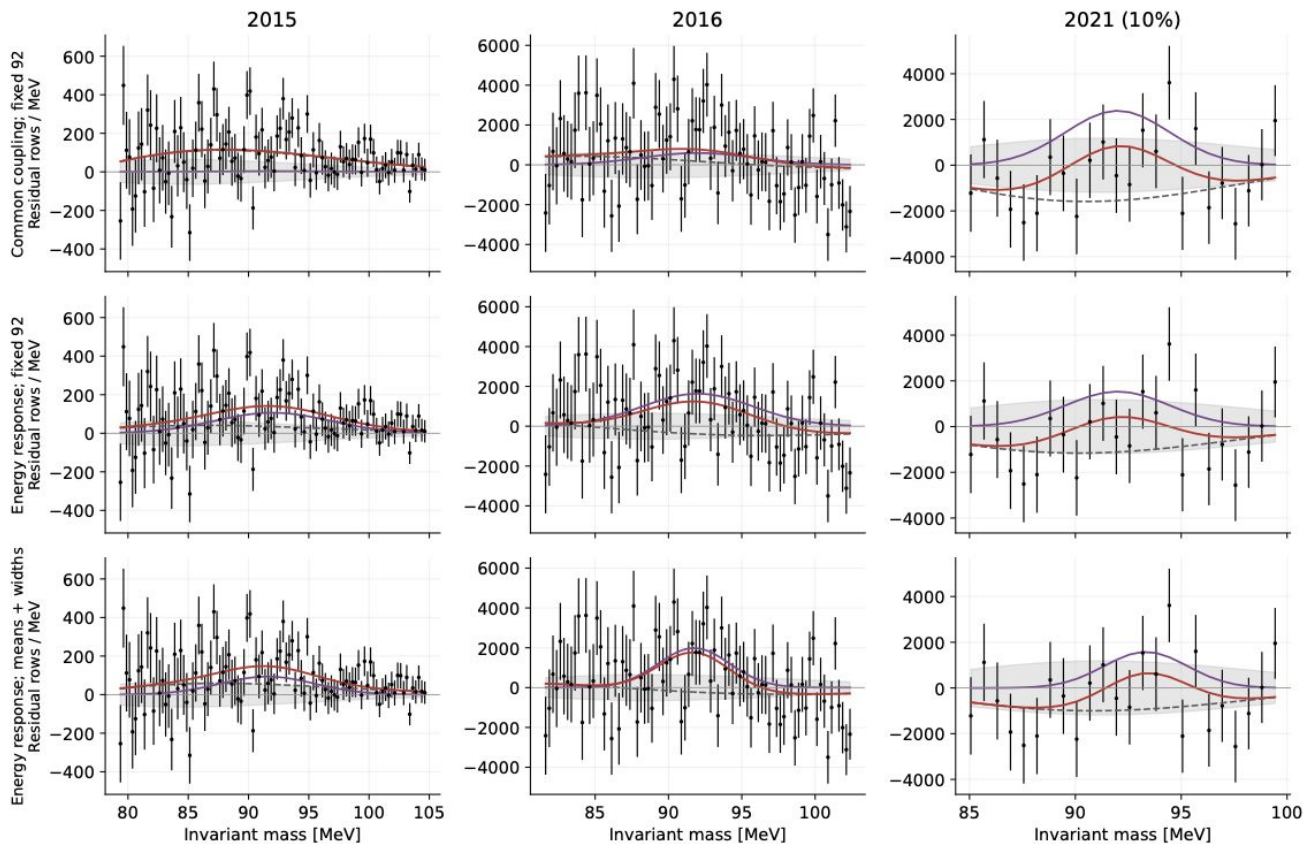


The Different Scenarios

Shared Coupling

Rate Floating
within Energy
Normalization

Rate Floating
within Energy
Normalization+
Mass Resolution
Profiled



Updates and Studies Since Initial Unblinding (08/04/2026)



- I. Geometric Acceptance Study for 2015/2016/2021
- II. Pseudoexperiments of various Statistics
 - A. 1% fit data scaled to 10% and 100%
 - B. 10% fit data fit as 10% and scaled to 100%
- III. 2021 GP fit support optimization
 - A. rising edge
- IV. New observed limit results and all combinations of limits
- V. 2016 100% Comparison
- VI. Contour revitalization (15/16/21 and 15/16/19/21 Projections)
- VII. GPR Echo characterization across all three datasets
 - A. individual/combined dataset search correlations
 - B. predictions for 100% unblinding
- VIII. Scaling Mass Resolution Study
- IX. Profiling Mass resolution as a nuisance
- X. Beam Energy relationship to extracted regions of interest
- XI. “Two-bump” hypothesis